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Chaotic dynamics of sea clutter

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The notion that a deterministic nonlinear dynamical system (with relatively few degrees of freedom) can display aperiodic behavior has a strong bearing on sea clutter characterization: random-looking sea clutter may be the outcome of a chaotic process. This new approach envisages deterministic rules for the underlying sea clutter dynamics, in contrast to the stochastic approach where sea clutter is viewed as a random process with a large number of degrees of freedom. In this paper, we demonstrate, convincingly for the first time, the chaotic dynamics of sea clutter. We say so on the basis of results obtained using radar data collected from a series of extensive and thorough experiments, which have been carried out with ground-truthed sea clutter data sets at three different sites. The study includes correlation dimension analysis (based on the maximum likelihood principle) and Lyapunov spectrum analysis. The Lyapunov (Kaplan-Yorke) dimension, which is a byproduct of Lyapunov spectrum analysis, shows that it is indeed a good estimator of the correlation dimension. The Lyapunov spectrum also reveals that sea clutter is produced by a coupled system of nonlinear differential equations of order five or six. © 1997 American Institute of Physics. [S1054-1500(97)01504-8]

Sea clutter refers to the backscattered returns from a patch of the sea surface illuminated by a transmitted radar pulse. The study of sea clutter is not only of theoretical importance but also of practical importance because it places severe limits on the detectability of radar returns from “point” targets (e.g., low-flying aircraft, small marine vessels, navigation buoys, and small pieces of ice) on or near the sea surface. In this paper, we present highly detailed experimental evidence for the chaotic dynamics of sea clutter. The experimental data used in the analysis were obtained using different radar systems operating at low grazing angles at different locations and under varying environmental conditions. Specifically, we show that the generation of sea clutter is indeed governed by a coupled system of five or six nonlinear differential equations, which resides on a geometric object (attractor) of dimension between 4.1 and 4.5.

I. INTRODUCTION

“Sea clutter,” referring to radar backscatter from sea surface, has a long history of being modelled as a stochastic process, which goes back to the early work of Goldstein.¹ One of the main reasons for this approach has been the random-looking behavior of the sea clutter waveform. In the classical view, going back to Boltzmann, the irregular behavior of a physical process encountered in nature is believed to be due to the interaction of a large number of degrees of freedom in the system. This has led to the conclusion, in general, that complex behavior is due to the characterization of a complex system with a large number of free parameters, hence the justification for the stochastic approach. Contrary

to this approach, recent developments in the theory of chaos and nonlinear dynamics have shown that many nonlinear and deterministic dynamical systems with relatively few degrees of freedom exhibit random/complex/irregular behavior in their dynamics.^{2–10} This, in turn, has opened up a powerful and elegant way of looking at complex behavior of physical processes, whereby they can be modelled as simple, nonlinear and deterministic dynamical systems. Here, the word simple means a system with few degrees of freedom. We are reminded here of the fact that one of the fundamental issues in the mathematical modelling of a physical process is that of minimizing the number of degrees of freedom required to describe the process.^{2–10}

With the emergence of powerful analytic methods for handling time series generated by nonlinear dynamics, a different set of descriptive measures for sea clutter is now available. Indeed, recent experimental results suggest that there may be a low-dimensional dynamical attractor controlling the behavior of sea clutter.^{11,12} Haykin and Leung were the first to claim the chaotic dynamics of sea clutter by looking at the fractal correlation dimension (D_2).^{13–15} They used the Grassberger-Procaccia algorithm (GPA)¹⁶ to estimate D_2 , and its value was reported to be between 6–9.^{13–15} In 1992, He and Haykin moved one step forward to verify the chaotic dynamics of sea clutter by calculating the largest Lyapunov exponent.¹⁷ They used Wolf’s method¹⁸ and showed that the largest Lyapunov exponent is positive. Later, Haykin and Li re-established the presence of a low-dimensional chaotic attractor for sea clutter dynamics and they made use of this property of sea clutter for the enhanced detection of a radar target embedded in sea clutter.^{19–22} Studies by Palmer *et al.* have also shown that sea clutter can be modelled as a low-dimensional attractor.²³

However, there has been a great deal of disagreement between the results on the chaotic characterization of sea

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clutter data reported by various researchers. For example, Leung and Haykin reported a D_2 value between 6 and 9 from their analysis using GPA.^{13–15} Haykin and Li reported a D_2 value between 7 and 9.²² Palmer *et al.* reported a D_2 value between 5 and 8.²³ Li reported one positive Lyapunov exponent followed by a zero exponent and several negative exponents;²² he also reported a Kaplan-Yorke dimension (D_{KY}) around 3, which is well below the estimated D_2 values. Ideally, the D_{KY} should be closer to the D_2 .^{24–29} These discrepancies are attributed to the types of algorithm used for estimating D_2 and Lyapunov exponents. The GPA for the estimation of D_2 and Wolf's algorithm for the estimation of the largest Lyapunov exponent require data with high signal-to-noise ratio (SNR) for the estimates to be reliable.^{16,18,30} These algorithms are highly sensitive to the presence of noise in the collected data and produce spurious estimates of these parameters if the SNR is below 20 dB.^{16,18}

These discrepancies and the availability of other more "robust" algorithms for estimating D_2 and the Lyapunov spectrum prompted the senior author of this paper to revisit the problem of chaotic characterization of sea clutter. In the experimental study reported in this paper, the various tasks of interest have been solved using specific algorithms as indicated here:

- (1) Test for nonlinearity in three different ways, using (i) a wave-shaping filter, (ii) surrogate data analysis based on the growth of inter-point distances as the discriminating statistic, and (iii) surrogate data analysis based on D_2 as the discriminating statistic, hereafter referred to as WSF, SIPD and SCD, respectively.^{31,32}
- (2) Correlation dimension estimation, using the maximum likelihood principle as described in Schouten *et al.*,³³ hereafter referred to as the STB algorithm after its originators.
- (3) Normalized embedding delay, using Shannon's mutual information criterion as described in Fraser and Swinney,^{34,35} hereafter referred to as the MI algorithm.
- (4) Global embedding dimension (d_E), using the method of false nearest neighbors described in Kennel *et al.*,³⁶ hereafter referred to as the GFNN algorithm.
- (5) Local embedding dimension (d_L), using the method of local false nearest neighbors described in Abarbanel and Kennel,³⁷ hereafter referred to as the LFNN algorithm.
- (6) Lyapunov spectrum, using the method described in Brown *et al.*, hereafter referred to as the BBA algorithm after its originators.^{38–40}

Detailed aspects of these algorithms are presented later. [Two specific pieces of software were used for the implementation of the algorithms: (1) RRCHAOS, developed at Delft University, the Netherlands, for the SIPD and STB algorithms. (2) cspX, developed by the Applied Chaos Group at Del Mar, California, for the MI, GFNN, LFNN, and BBA algorithms.] The results presented in this paper, obtained with these algorithms, demonstrate the chaotic dynamics of sea clutter in a very convincing way. The results

were obtained using ground-truthed radar data collected from a series of extensive and thorough experiments which were carried out on the East Coast of Canada.

The paper is organized as follows. In Section II we present a brief review of sea clutter. In Section III we describe the different radar environments used for the collection of sea clutter data. In Section IV we describe a list of criteria for assessing the chaotic dynamics of an experimental time series. In Section V we describe the various forms of preprocessing used to prepare the data for chaotic analysis. In Section VI we describe tests for continuity and differentiability performed on a filtered time series. In Section VII we describe tests for stationarity and nonlinearity of an experimental time series. In Section VIII we describe the algorithms for the estimation of chaotic invariants of a nonlinear dynamical system using experimental data. In Section IX we present the results of these tests and estimation algorithms applied to real-life sea clutter data. The paper concludes with some final remarks in Section X.

II. SEA CLUTTER

Sea clutter or *sea echo* refers to the backscattered returns from a patch of sea surface illuminated by a transmitted radar signal. The study of sea clutter is not only of theoretical but also of practical importance because it places severe limits on the detectability of radar returns from "point" targets on or near the surface. Such targets include low-flying aircraft, small marine vessels, navigation buoys, and pieces of ice floating in the ocean.

Our interest in this paper is confined to the underlying dynamics of sea clutter acting on its own. More specifically, we are interested in sea clutter from a microwave radar operating at low grazing angles with the antenna dwelling in a fixed direction. In such a mode of operation, the *temporal* character of sea clutter is entirely due to the motion of the sea surface. This particular mode of operation is chosen in order to focus on the interaction between the incident electromagnetic wave and the sea surface solely as a function of time and in the simplest manner possible. (In a scanning mode of operation as in surveillance radar, the received signal assumes a more complicated form in that it is a function of both time and space.)

The illuminated area of sea surface responsible for the radar backscatter (i.e., the *footprint* of the antenna) is determined by three factors:

- (1) Beamwidth of the antenna pattern.
- (2) Height of the antenna above the sea surface.
- (3) Nominal grazing angle of the incident electromagnetic wave (measured with respect to the sea surface), which is usually defined at the beam center (boresight).

The antenna footprint may be sub-divided in range into *resolution cells*, as illustrated in Fig. 1 for a pencil (cone-shaped) beam. The cross-range resolution of each cell is determined by the beamwidth of the antenna. The range resolution, denoted by ΔR , is defined as the minimum distance in range by

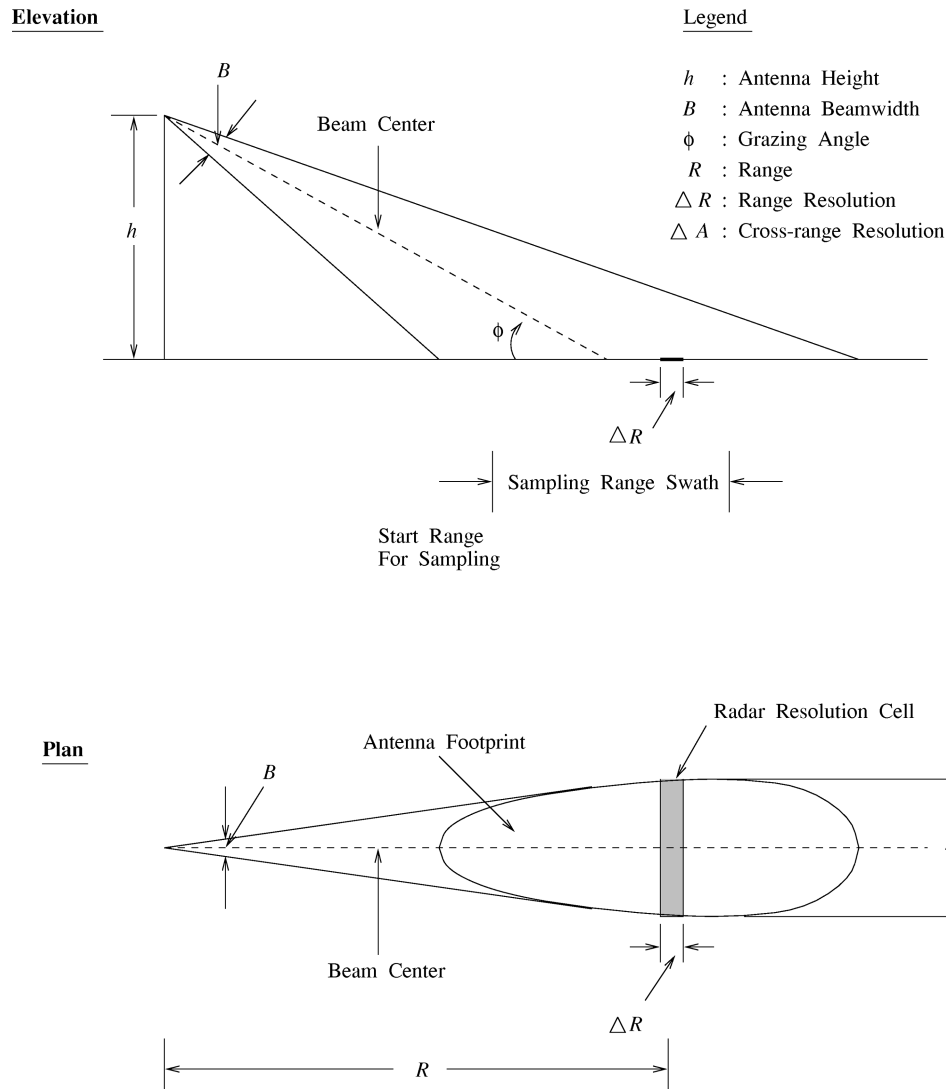


FIG. 1. Elevation and plan views of the antenna footprint illuminated by a radar, and the sampling area from which data are collected. Note: The grazing angle and antenna beamwidth shown in this figure are larger than the actual values in the radar experiments; this has been done here merely for the purpose of illustration.

which two point targets must be separated in order to be individually resolved. It is determined by the duration of the transmitted radar pulse, t_0 , as shown by

$$\Delta R = \frac{ct_0}{2}, \quad (1)$$

where c is the velocity of propagation. In the case of a radar using pulse compression, t_0 is the “effective” pulse duration. Thus, for a particular range (i.e., distance from the radar), the backscattered return is integrated over the area of the pertinent resolution cell. Note that the resolution cells are of unequal sizes in azimuth due to the spreading (with range) of the antenna footprint, as illustrated in Fig. 1.

In addition to the above-mentioned factors, the characterization of sea clutter depends on the following:

- (i) **Air-sea interactions**, which are themselves influenced by the direction and speed of prevalent winds,

and also by the wind field at distant locations that generate waves that eventually propagate to the local area.

- (ii) **Radar parameters**, namely, transmitted pulse duration, wavelength, and polarization on transmit and receive.

Figure 2 shows sample time histories of the received signal amplitude versus range for five different situations. Figure 2(a) shows the amplitude plot for radar bearing angle of 30° and horizontal polarization on both transmit and receive (hereafter referred to as horizontal like-polarization or HH). There was no wind, and the wave height was 0.85 m. As expected, there is no particular pattern seen in this first amplitude plot. Figure 2(b) shows the amplitude plot for radar bearing angle of 105° and vertical polarization on both transmit and receive (hereafter referred to as vertical like-polarization or VV). The winds were 19 kilometers/hour

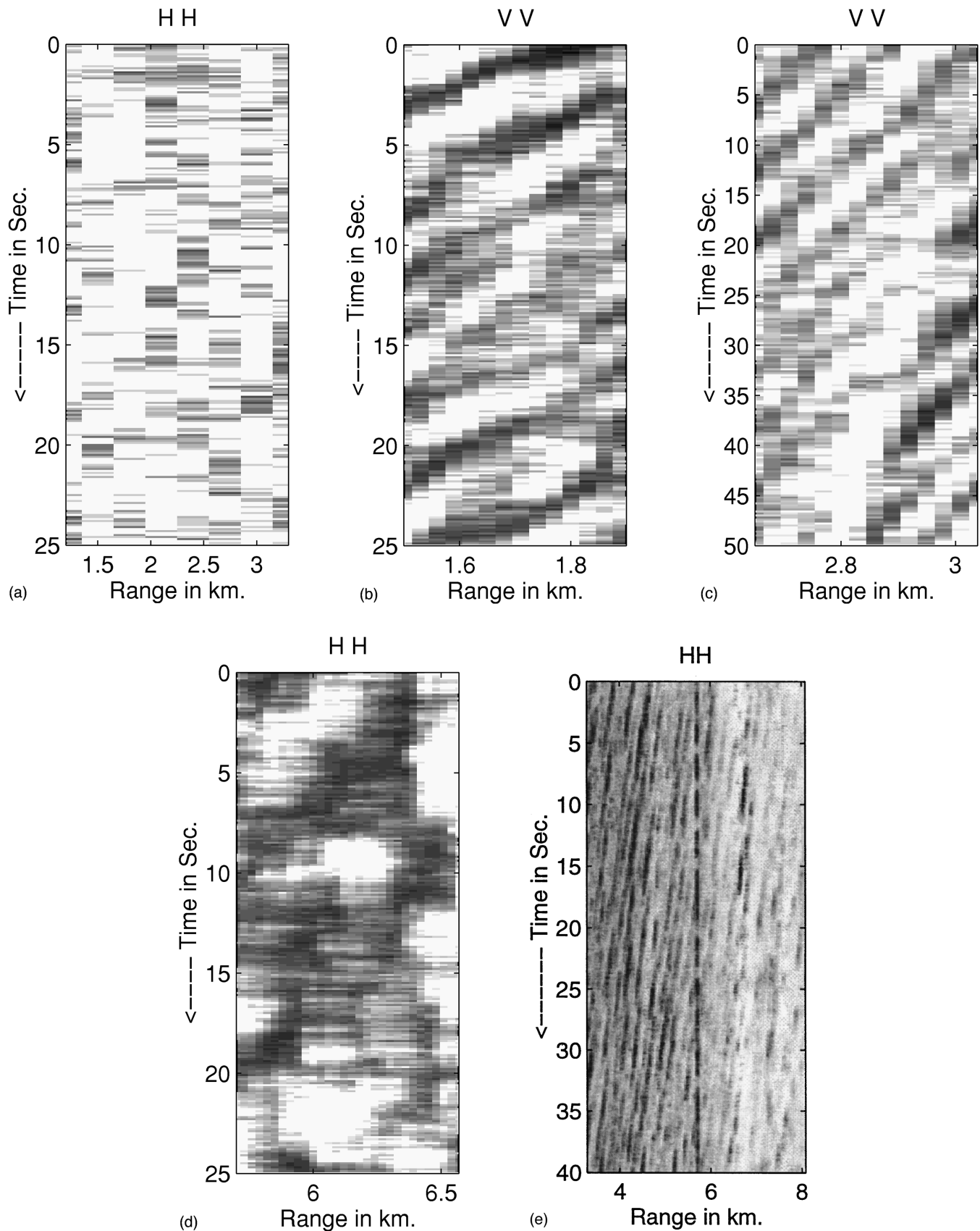
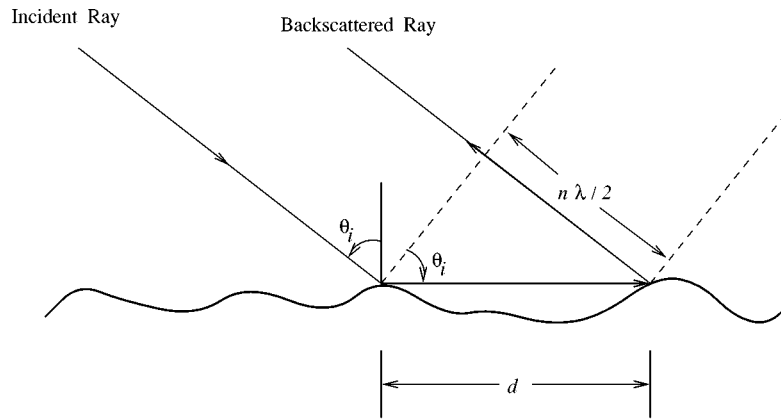


FIG. 2. Time histories of received signal amplitude versus range for five different situations: (a) $\Delta R=300.0$ m, range sampling rate=5 MHz, wave height=0.85 m. (b) $\Delta R=30.0$ m, range sampling rate=10 MHz, wave height=1.3 m. (c) $\Delta R=30.0$ m, range sampling rate=10 MHz, wave height=2.6 m. (d) $\Delta R=30.0$ m, range sampling rate=30 MHz, wave height=2.5 m. (e) $\Delta R=37.5$ m, range sampling rate=8 MHz, wave height=2.0 m. (For more details see Section III).



Legend

- θ_i : Incidence Angle
 n : Grating Order
 d : Surface Wavelength
 λ : Radar Wavelength

FIG. 3. Bragg scattering condition from a rough sea surface.

from 90° , and the wave height was about 1.3 m. The swell waves are clearly visible in this second figure: an expected result as the radar was looking closely into the swell and the wind direction. Figure 2(c) shows the amplitude plot for radar bearing angle of 190° and vertical like-polarization. The winds were 26 kilometers/hour from 320° . This third situation shows some indication of the swell waves as the radar is looking essentially downwind. Figure 2(d) shows the amplitude plot for radar bearing angle of 30° and horizontal like-polarization. The winds were 31 kilometers/hour from 120° , and the wave height was 2.5 m. This fourth figure shows no particular wave pattern: again an expected result as the radar was looking essentially cross-wind and cross-swell. Finally, Fig. 2(e) shows the amplitude plot for radar bearing angle of 270° and horizontal like-polarization. The winds were 36 kilometers/hour from 190° , and the wave height was 2.0 m. The swell waves are clearly visible in Fig. 2(e). Note, however, this latter figure is on a much larger range scale than that of Fig. 2(b), hence the difference in their appearance. The continuous response at the center range is from a moored buoy.

A. Sea clutter at low grazing angles

Basically, there are two types of surface waves that influence radar backscatter at low grazing angles; they are capillary waves and gravity waves, as described here:⁴¹

- (i) **Capillary waves** provide the fine structure of the sea surface. Their wavelengths are on the order of two centimeters or less. The dominant restoring force for capillary waves is surface tension.

- (ii) **Gravity waves** provide the larger and more visible structure of the sea surface. Their wavelengths vary from 200 meters down to a small fraction of a meter. The dominant restoring force for gravity waves is the force of gravity, hence their name. Gravity waves may reside in one of two states: sea and swell. *Sea* is the state of waves when they are blown by the wind that gave rise to them. *Swell* is the state of waves when they are no longer under the influence of the wind that generated them. Since the surface over which the waves travel acts as a low-pass filter, swell waves often take the form of long-crested low-frequency sinusoids.

In a microwave radar, the wavelength is on the order of a few centimeters or less. For example, the wavelength of an X-band radar (the type of the radar used in the study reported in this paper), the wavelength is about 3.0 cm. For a simplified model of sea clutter produced by a microwave radar operating at low grazing angles, we may thus consider resonant or *Bragg scatter* from the large number of small capillary waves riding on top of the gravity waves. What is being described here is a two-scale backscatter model,^{42,43} in which the surface waves act as a kind of diffraction grating, as illustrated in Fig. 3. From the grating equation of optics, the condition for *constructive* interference in the backscatter direction is defined by

$$n \frac{\lambda}{2} = d \sin \theta_i, \quad (2)$$

where n is an integer, λ is the radar wavelength, d is the

surface wavelength, and θ_i is the incidence angle (measured with respect to the normal to the sea surface); that is, θ_i is the complement of the grazing angle. For a low grazing angle, θ_i is close to 90° and $\sin \theta_i \approx 1$. Thus, the capillary waves causing first order ($n=1$) Bragg scattering are those waves whose wavelengths are about one-half the radar wavelength.

The simplified Bragg scatter model, described above, ignores shadowing and diffraction effects that do arise at low grazing angles. These effects make the relationship between the scattered field and the wave-height profile of the sea surface highly nonlinear. The net result is that sea clutter is a nonlinear physical process.

B. Compound K-distributions

Traditionally, sea clutter has been modelled as a stochastic (random) process, with the observable received signal from a radar resolution cell representing a sample function of that process. This approach was motivated by the classic paper of Rice⁴⁴ on Gaussian noise theory. Longuet-Higgins⁴⁵ generalized Rice's results to spatial as well as temporal variations in the modelling of rough surfaces.

When the pulse duration of a microwave radar is sufficiently short, the amplitude distribution of sea clutter has an extended tail, compared to the Rayleigh distribution that follows from Gaussian noise theory. This deviation was first recognized by Goldstein.¹ In an effort to provide better fits to the amplitude distribution of sea clutter, *ad hoc* distributions, namely, log-normal⁴⁶ and Weibull,⁴⁷ were proposed during the 1970s. About the same time, non-Gaussian light experiments performed on layers of liquid crystals in turbulent motion were found to generate signals with statistical properties remarkably similar to those of sea clutter. It was then realized that the two-dimensional random-walk model⁴⁸ that was being developed to explain those non-Gaussian light experiments could also form the statistical basis for addressing the sea clutter problem. Indeed, Jakeman and Pusey⁴⁹ were the first to propose K-distributions, based on a modified Bessel function of the second kind, as a statistical model for sea clutter data. Among the various non-Gaussian models reported in the literature on the statistical characterization of scattered waves from rough surfaces, the K-distribution model stands out for a combination of two reasons: (i) phenomenological arguments, and (ii) exact solutions to specific scattering problems. Jakeman and Tough⁵⁰ develop a highly detailed statistical mechanical formulation for non-Gaussian compound Markov processes, with K-distributed noise treated as a special case.

In a physical sense, randomness is introduced into the K-distribution model for a continuous-medium scattering problem by invoking the following phenomenological arguments:

- (i) The problem is formulated as one of scattering by a sub-fractal random-phase changing screen, which is defined as a diffusing layer that introduces spatially random variables in phase into the incident plane wave.⁵¹
- (ii) As the distorted wavefront propagates beyond the dif-

fusing layer, amplitude fluctuations are produced with characteristics governed by the statistical and correlation properties of that layer.^{52,53}

An intuitively appealing way of looking at the K-distribution is as a Rayleigh process with its scale parameter varying as a gamma-distributed variate, which may reflect the Bragg scatter being modulated by the large-scale gravity waves. Another appealing feature of K-distributions is the fairly close fit they provide for the amplitude fluctuations of real-life sea clutter data over a wide range of experimental conditions.⁵⁴ However, for very high-resolution radars the K-distribution model fails, hence the current research efforts to use higher compound models or spherically invariant processes.⁵⁵ In any event, the major limitation of the K-distribution model or extensions thereof is the fact that none of these models says anything about the underlying dynamics responsible for the generation of sea clutter. Information about the dynamics of sea clutter is particularly important in radar signal processing. For example, by using a nonlinear predictive model designed to capture the underlying dynamics of sea clutter, the target signal-to-clutter ratio at the output of the predictive model is made much larger than the corresponding value at the model input, and detectability of the target in sea clutter is thereby significantly improved.¹⁹

III. RADAR AND DATABASE

A large part of the database used for the study was collected using an instrument-quality radar called the IPIX radar.⁵⁶ The IPIX radar has the following features. It is dual-polarized and coherent, with a beamwidth of 1° . It has built-in calibration equipment, which permits quantitative measurements to be made with high confidence, and a digital control system which allows the experimental parameters to be easily varied. The radar was mounted in a fixed position on land, 25–30 m above sea level. This height is typical of a ship-mounted radar. It was operated in a dwelling mode, with the antenna pointing along a fixed direction, illuminating a patch of the ocean surface.

Sea clutter data were recorded from different resolution cells, defined within the antenna footprint (see Fig. 1). The start range values for sampling within the antenna footprint were 1200 m or greater for different experiments, and the range sampling rate varied from 5 MHz to 30 MHz with an 8-bit A/D converter.

- (i) *Experiments at Cape Bonavista, Newfoundland:* The IPIX radar system was first installed at Cape Bonavista, Newfoundland, Canada, in May–June 1989. A wide variety of sea states can be observed at that site throughout the year. Sea clutter data for various sea states were collected. For each data set, I-channel (in-phase) and Q-channel (quadrature-phase) recordings were made, each consisting of a sampled range swath for at least 50 000 consecutive sweeps.
- (ii) *Experiments at Dartmouth, Nova Scotia:* Further field experiments at Dartmouth, Nova Scotia, Canada, were carried out in November 1993, using a refined version

of the IPIX radar. Sea clutter (I and Q) data for various sea states, radar pulse durations, pulse repetition frequencies (PRF), and different like-polarizations were recorded.

- (iii) *Experiments at Argentina, Newfoundland:* For these experiments, a noncoherent commercial marine radar system was installed at a Vessel Traffic System (VTS) site in Argentina, Newfoundland, in January-February 1994. The radar data collected at this site were confined to amplitude information only for fixed radar parameters.

Five sea clutter data sets, collected from these radar experiments at different times of the day, different months of the year, and under different environmental conditions, were analyzed using the tools described later. The radar parameters and environmental conditions for the data sets were as follows:

1. Data Set I (Dartmouth)

Date	:	17/11/1993
Time	:	7:57
Radar bearing angle	:	30°
Polarization	:	HH
Pulse Repetition Frequency	:	2000 Hz
Pulse Duration	:	2000 ns
Beamwidth	:	1°
Range Resolution	:	300.0 m
Area of First Resolution Cell	:	≈300.0 m×3.33 m
Range Sampling Rate	:	5 MHz
Start Range	:	1.2 km
Significant Wave Height	:	0.85 m
Wind	:	None

2. Data Set II (Dartmouth)

Date	:	14/11/1993
Time	:	15:13
Radar bearing angle	:	105°
Polarization	:	VV
Pulse Repetition Frequency	:	2000 Hz
Pulse Duration	:	200 ns
Beamwidth	:	1°
Range Resolution	:	30.0 m
Area of First Resolution Cell	:	≈30.0 m×1.47 m
Range Sampling Rate	:	10 MHz
Start Range	:	1.5 km
Significant Wave Height	:	1.3 m
Wind	:	19.0 km/h from 90°

3. Data Set III (Dartmouth)

Date	:	17/11/1993
Time	:	21:38
Radar bearing angle	:	190°
Polarization	:	VV
Pulse Repetition Frequency	:	1000 Hz
Pulse Duration	:	200 ns
Beamwidth	:	1°
Range Resolution	:	30.0 m
Area of First Resolution Cell	:	≈30.0 m×7.22 m
Range Sampling Rate	:	10 MHz

Start Range	:	2.6 km
Significant Wave Height	:	2.6 m
Wind	:	26.0 km/h from 320°

4. Data Set IV (Cape Bonavista)

Date	:	12/06/1989
Time	:	13:33
Radar bearing angle	:	30°
Polarization	:	HH
Pulse Repetition Frequency	:	2000 Hz
Pulse Duration	:	200 ns
Beamwidth	:	1°
Range Resolution	:	30.0 m
Area of First Resolution Cell	:	≈30.0 m×15.83 m
Range Sampling Rate	:	30 MHz
Start Range	:	5.7 km
Significant Wave Height	:	2.5 m
Wind	:	31.0 km/h from 120°

5. Data Set V (Argentina)

Date	:	08/02/1994
Time	:	4:26
Radar bearing angle	:	270°
Polarization	:	HH
Pulse Repetition Frequency	:	1300 Hz
Pulse Duration	:	250 ns
Beamwidth	:	1.3°
Range Resolution	:	37.5 m
Area of First Resolution Cell	:	≈37.5 m×11.92 m
Range Sampling Rate	:	8 MHz
Start Range	:	3.3 km
Significant Wave Height	:	2.0 m
Wind	:	36.0 km/h from 190°

The time histories of amplitude versus range for these five data sets are displayed in Figs. 2(a) to 2(e), respectively.

IV. CRITERIA FOR ASSESSING THE CHAOTIC DYNAMICS OF A PHYSICAL TIME SERIES

Given a physical (experimental) time series, how do we determine whether the time series is derived from a chaotic process? Unfortunately, there is no single criterion that we may invoke to answer this important question. Rather, we have a set of criteria, all of which must be satisfied if we are to be certain that the given time series is the result of a chaotic process. These criteria are summarized here:

- (1) The process should be nonlinear.
- (2) The attractor (correlation) dimension (D_2) of the process should be fractal. Moreover, D_2 should converge to a constant value for increasing embedding dimension.
- (3) The dynamics of the system responsible for the generation of the process should be sensitive to initial conditions. This, in turn, means that at least one of the Lyapunov exponents of the process should be positive. The largest Lyapunov exponent determines the horizon within which the time series is predictable.

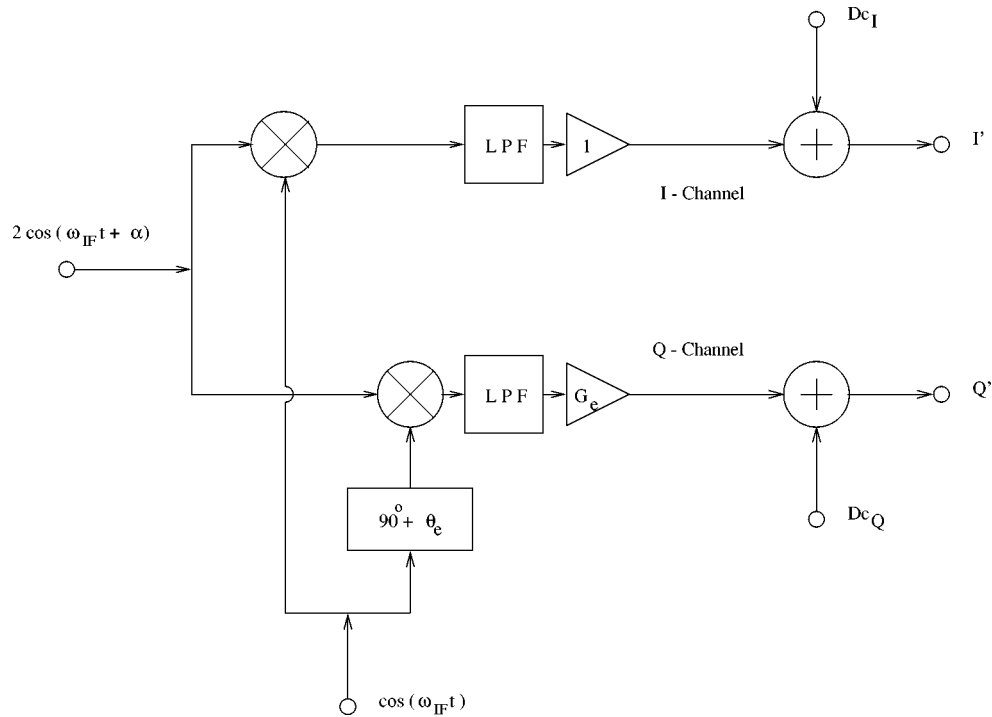


FIG. 4. Circuit model for I-Q calibration.

- (4) The sum of all Lyapunov exponents of the process should be negative for the underlying dynamics to be dissipative (i.e., physically realizable).
- (5) The Kaplan-Yorke dimension (D_{KY}), a byproduct of the Lyapunov spectrum, should be close in numerical value to the correlation dimension.

The time series should be long enough for the estimates of the correlation dimension and Lyapunov spectrum to be reliable. However, the time series should be stationary for the estimation algorithms to function properly.

One other point that needs attention is that a physical time series is typically contaminated with noise. It may therefore be necessary to filter the time series before proceeding to ascertain its chaotic dynamics. To do so, however, care has to be taken to ensure that the chaotic dynamics of the process are not affected by the filtering operation.

V. PREPROCESSING OF THE EXPERIMENTAL DATA

Preprocessing of the measured data is an essential step in any experimental data analysis. The experimental data collected from different sites using different radars were influenced by many types of errors: (i) receiver noise, (ii) quantization error due to the A/D converter, and (iii) in-phase and quadrature-phase component misalignment due to gain and phase imbalances. These errors were minimized by suitable preprocessing of the data, before proceeding to extract the chaotic parameters. Most importantly, however, extra care was taken to ensure that the preprocessing did not alter the dynamics of the underlying clutter process. In this study, the following preprocessing techniques were employed on the radar data, in the order presented here.

A. Amplitude and phase corrections (I-Q calibration)

The raw in-phase and quadrature-phase components of the coherent radar showed amplitude and phase mismatches (unequal variance and nonzero cross-correlation) because of experimental errors. These were due to direct current (dc) offsets and differences in the gain and phase characteristics of the individual receivers. These errors were corrected by I-Q calibration method, as detailed below.

The I-Q calibration procedure was applied to measure the gain and phase imbalances between the I and Q channels. Figure 4 shows the circuit diagram for doing this calibration. The low-pass filters included in this figure were designed to remove the high-frequency components appearing at the multiplier outputs. The receiver channel gains for both I and Q channels were normalized by using the I channel as reference. This resulted in a unity gain for the I channel and G_e for the Q channel. θ_e represents the relative phase error in the Q channel with respect to the I channel. Dc_I and Dc_Q represent the dc offsets in I and Q channels, respectively. If the input is an IF signal of the form $2 \cos(\omega_{IF}t + \alpha)$, the outputs of the I and Q channels would ideally be as follows (assuming no errors):

$$I = A \cos(\alpha) \quad (3)$$

and

$$Q = A \sin(\alpha) \quad (4)$$

where A is a scaling factor. However, because of unavoidable gain and phase errors, as well as the dc offsets in the two channels, the actual outputs in Fig. 4 are

$$I' = A \cos(\alpha) + Dc_I \quad (5)$$

and

$$Q' = AG_e \sin(\alpha - \theta_e) + Dc_Q. \quad (6)$$

Now, solving for $\cos(\alpha)$ and $\sin(\alpha)$ from equations (5) and (6) and substituting them into equations (3) and (4), we obtain the actual I and Q components of the signal as follows:

$$I = I' - Dc_I \quad (7)$$

and

$$Q = \frac{1}{\cos(\theta_e)} \left[\frac{Q' - Dc_Q}{G_e} - I \sin(\theta_e) \right]. \quad (8)$$

The parameters G_e and θ_e for the recorded I and Q values (for a data length N) are computed using the following formulas:

$$G_e = \left[\frac{\sum_{i=1}^N (Q'(i) - Dc_Q)^2}{\sum_{i=1}^N (I'(i) - Dc_I)^2} \right]^{1/2} \quad (9)$$

and

$$\theta_e = \sin^{-1} \left[\frac{2 \sum_{i=1}^N (I'(i) - Dc_I) [(Q'(i) - Dc_Q)/G_e]}{\sum_{i=1}^N (I'(i) - Dc_I)^2 + [(Q'(i) - Dc_Q)/G_e]^2} \right]. \quad (10)$$

This calibration process was applied to the sea clutter data from Cape Bonavista and Dartmouth, where I and Q channels of data are available separately. In each case, the amplitude of the clutter data was computed in the following way (after performing filtering operations on I-Q corrected I and Q components, independently):

$$A = \sqrt{I^2 + Q^2}. \quad (11)$$

As for the Argentina data sets, there is no need for such a correction to be made as the radar system has a single component, namely, amplitude.

B. Filtering

The collected data may also be influenced by measurement noise (quantization error and receiver noise) or additive noise (artifacts). These artifacts must be minimized by applying suitable filters to the recorded signal. Here again, care must be taken in choosing the smoothing filters to make sure that the filtering does not affect the underlying dynamics of the system which produced the time series.⁵⁷⁻⁶² We used two kinds of filtering to reduce the effect of experimental noise. In the first method, we used a smoothing filter of the form $\hat{x}(n) = \sum_{i=1}^3 a_i x(n+i-2)$ with $a_i = 1/3$ to smooth the raw clutter data. In the second method, we designed a low-pass linear phase, finite impulse response (FIR) filter having 100 coefficients to filter the raw clutter data. While designing the filter, care was taken to ensure that the bandwidth of the filter was large enough so as not to affect the clutter data. This issue is discussed in the next section.

VI. CONTINUITY AND DIFFERENTIABILITY TESTS ON FILTERED DATA

We are interested in characterizing, in a statistical sense, the mathematical properties of embedding from one time series to second time series. By embedding we mean the reconstruction of an unknown dynamical system by formulating a new state-space using successive values of the observable time series. In our case, the first time series is the unfiltered sea clutter data and the second time series is the filtered sea clutter data. The essence of this mathematical analysis is to show that the trajectories formed from these two time series are diffeomorphically related to each other. The diffeomorphic relationship between these two time series is analyzed by testing for *continuity* and *differentiability*,⁶³ which are important properties pertaining to the preservation of the link between dynamics of the two time series.

A. Continuity

Let $\mathbf{X}$ be the phase-space constituted by the first time series and $\mathbf{Y}$ be the phase-space constituted by the second time series. Let f be the mapping (linear filtering in our case) which generates the second time series from the first. The map f is said to be continuous if there are no points in the phase-space $\mathbf{X}$ that are arbitrarily close, but that are mapped to distant points on the phase-space $\mathbf{Y}$. Putting it in a mathematical form, the function f is continuous if the following condition is satisfied:

$$\text{At a point } \mathbf{x}_0 \in \mathbf{X} \text{ if } \forall \varepsilon > 0 \exists \delta > 0 \text{ such that } \|\mathbf{x} - \mathbf{x}_0\| < \delta \Rightarrow \|f(\mathbf{x}) - f(\mathbf{x}_0)\| < \varepsilon.$$

We define a quantity called the *continuity index*, denoted by $\Theta_{C^0}(\varepsilon)$, to statistically test the continuity between the two time series embeddings. When $\Theta_{C^0}(\varepsilon) \approx 1$, then we can be confident that the mapping is continuous by rejecting the null hypothesis that the points in $\mathbf{X}$ and $\mathbf{Y}$ are randomly related.

B. Differentiability

This property tells us about the smoothness of the function f . A mathematical definition for differentiability is as follows:

The mapping $f: \mathbf{X} \rightarrow \mathbf{Y}$ is said to be differentiable at a point $\mathbf{x}_0 \in \mathbf{X}$ if there exists a linear operator $\mathbf{A}$ such that $\forall \varepsilon > 0 \exists \delta > 0$, for which $\|\mathbf{x} - \mathbf{x}_0\| < \delta \Rightarrow \|f(\mathbf{x}_0) + \mathbf{A}(\mathbf{x} - \mathbf{x}_0) - f(\mathbf{x})\| < \varepsilon \|\mathbf{x} - \mathbf{x}_0\|$.

This means that f is well approximated locally by $\mathbf{A}$ and is therefore smooth. We have chosen to use the well-known statistic of multivariate correlation and use a null hypothesis that generates a probability for confidence statistic analogous to that for the continuity at $\mathbf{x}_0$. Here, we define another quantity called the *differentiability index*, denoted by $\Theta_{C^1}(\varepsilon)$, to statistically test for smoothness of the mapping. When $\Theta_{C^1}(\varepsilon) \approx 1$, then we can be confident that the mapping is differentiable (by rejecting the null hypothesis).

VII. PRELIMINARY TESTS

Before proceeding with the extraction of chaotic invariants from an experimental time series, it is advisable to test for the stationarity and nonlinearity of the time series, as described next.

A. Test of stationarity (recurrence plots)

The dynamical analysis of an experimental time series requires it to be drawn from a stationary process. To satisfy this requirement, we may use the so-called recurrence plot due to Eckmann *et al.*⁶⁴ This is a graphical tool for the diagnosis of drift and hidden periodicities in the time evolution of dynamical systems, which are unnoticeable otherwise. A brief description of construction of recurrence plot is given in the following paragraph.

Let $\mathbf{s}(i)$ be the i th point on the orbit describing a dynamical system in d_E -dimensional space. The recurrence plot is an array of dots in an $N \times N$ square, where a dot is placed at (i, j) whenever $\mathbf{s}(j)$ is sufficiently close to $\mathbf{s}(i)$. To obtain a recurrence plot from a time series $s(n)$, in practice, we may proceed as follows. First, choosing an embedding dimension d_E , we construct the d_E -dimensional orbit of $\mathbf{s}(i)$ by the method of delays (described later). Next, we choose $r(i)$ such that the ball of radius $r(i)$ centered at $\mathbf{s}(i)$ in $\mathbf{R}^{d_E}$ contains a reasonable number of other points $\mathbf{s}(j)$ of the orbit. Finally, we plot a dot at each point (i, j) for which $\mathbf{s}(j)$ is in the ball of radius $r(i)$ centered at $\mathbf{s}(i)$. This picture is called a *recurrence plot*. Note that the i and j are measured in units of time. These plots tend to be fairly symmetric with respect to the diagonal $i=j$, because if $\mathbf{s}(i)$ is close to $\mathbf{s}(j)$, then $\mathbf{s}(j)$ is close to $\mathbf{s}(i)$. We cannot expect to get complete symmetry of the pattern because of the fact that $r(i)$ and $r(j)$ may not be equal.

B. Tests of nonlinearity

As mentioned previously, nonlinearity is one of the requirements for a process to be chaotic. We may check for the nonlinearity of an experimental time series in one of three different ways:

- (1) using a wave-shaping filter (WSF),
- (2) using surrogate data analysis, with the growth of inter-point distances as the discriminating statistic³¹ (SIPD), and
- (3) using surrogate data analysis, with the correlation dimension D_2 as the discriminating statistic³² (SCD).

These methods are described in turn.

WSF Method: This first method perhaps gives the most convincing proof that a given time series is indeed a nonlinear process. First, a wave-shaping filter is used to generate a linear process with identically the same Fourier spectrum as that of the time series, as illustrated in Fig. 5. Let $S_x(\omega)$ be the power spectrum of the given time series, $x(n)$. A linear-phase finite impulse response (FIR) filter is designed in such a way that, when it is stimulated by a zero-mean white Gaussian noise (a linear process), it produces an output signal (a linear process) that has the same Fourier spectrum as

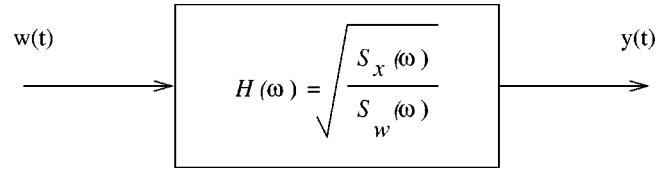


FIG. 5. Block diagram of the wave-shaping filter.

that of the original time series. [For the filter output to be truly stochastic, the stimulus must be stochastic too. This would therefore rule out the use of a pseudo-noise sequence generated on a computer, which, in reality, is a deterministic process. To satisfy the requirement that the stimulus in Fig. 5 be stochastic, the pure white Gaussian noise was generated using a commercially available analog noise generator (NC 1107A-1, Noise Com Inc.) coupled with an amplifier and A/D converter. This physical device contains a hermetically-packaged noise diode that has been burned-in for 168 hours and operates in a temperature range of -35 to $+100$ °C. It produces white Gaussian noise at $+13$ dBm and in a frequency range between 100 Hz and 100 MHz.] That is, the filter's magnitude response is of the form

$$H(\omega) = \sqrt{\frac{S_x(\omega)}{S_w(\omega)}}, \quad (12)$$

where $S_w(\omega)$ is the power spectrum of the white Gaussian noise. The filter coefficients are obtained by computing the inverse Fourier transform of $H(\omega)$, assuming a linear phase.

These two signals (i.e., the given time series and the output of the wave-shaping filter) are then subjected to the tests described in Section VIII. Those tests should reveal that the output of the wave-shaping filter is drawn from a random process (a known fact) and the given time series is different.

SIPD Method: In this second method,³¹ we perform a statistical test in which the given time series is compared with surrogate data sets. The surrogate data are created using a stochastic linear model with the same autocorrelation coefficients as those of the given time series. The (exponential) growth of inter-point distances is used as the discriminating statistic to test the null hypothesis that the original time series can be described by linearly correlated noise. A quantity Z is calculated from the Mann-Whitney rank-sum statistic. It is normally distributed with zero mean and unit variance under the null hypothesis that two observed samples of inter-point distances in the original and surrogate data sets come from the same population. A value of Z less than -3.0 is considered to be a valid reason for strong rejection.⁶⁵

SCD Method: This third and final method involves applying the surrogate data analysis proposed by Theiler *et al.*³² Surrogate data sets are generated and then compared with the original data set using the correlation dimension D_2 as the discriminating statistic. First, the Fourier spectrum of the original data set is computed for positive and negative frequencies (no windowing). This spectrum has a complex amplitude at each frequency. Next, the phase is randomized by multiplying each complex amplitude by $e^{j\phi}$, where ϕ is

picked from a uniform distribution in the interval $[0, 2\pi]$. In order for the inverse Fourier transform to be real, the phase is symmetrized by having $\phi(f) = \phi(-f)$. Finally, the inverse Fourier transform of the phase-randomized signal is computed to produce the surrogate data set. The results on D_2 analysis of the original time series and the corresponding surrogate data set should reveal that they are statistically distinct, thus again rejecting the null hypothesis that the original time series is produced by a linear stochastic process.

Having satisfied ourselves that the given experimental time series is indeed drawn from a process that is both stationary and nonlinear, then we proceed with the important task of extracting chaotic invariants from it.

VIII. EXTRACTION OF CHAOTIC INVARIANTS FROM EXPERIMENTAL DATA

Two important characteristic parameters of a chaotic process are (1) **correlation dimension** or **fractal dimension** (D_2), and (2) **Lyapunov exponents** (λ_i) and hence the **Lyapunov dimension** or Kaplan-Yorke dimension (D_{KY}).

It is an important concept in dynamics that dissipative dynamical systems are typically characterized by the presence of attracting sets or attractors.²⁻¹⁰ Thus, the trajectories of a dynamical system converge towards a limit set, an attractor, in the state space, which only fills a subset of the space. Because of the assumptions of boundedness and non-periodicity, the attractor necessarily has a very peculiar and intricate structure with possibly fractal properties, a situation referred to as strange attractor (footprints of chaos).²⁻¹⁰ The existence of a fractal dimension for an attractor can therefore be used as an indication of possibly chaotic dynamics of the underlying physical system, with the dimension defining a lower bound on the number of degrees of freedom required to characterize the system. Another consequence of the lack of periodicity is the broad-band spectrum of a chaotic signal, which is a necessary condition for the assessment of chaos. Thus, the chaotic characterization of a dynamical system can be achieved by studying the attractor of the system in a multidimensional state-space.⁷ A classical approach to the estimation of D_2 was provided by Grassberger and Procaccia.¹⁶ Their algorithm, popularly known as GPA, computes the number of couples $(\mathbf{s}(i), \mathbf{s}(j))$ whose magnitude is less than a given radius r in the multidimensional space.¹⁶ More precisely, GPA computes the correlation integral $C(r)$ given by

$$C(r) = \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \theta(r - \|\mathbf{s}(i) - \mathbf{s}(j)\|), \quad (13)$$

where θ is the Heaviside function, $\mathbf{s}(i)$ and $\mathbf{s}(j)$ are vectors in the multidimensional space, and N is the total number of vectors in the multidimensional space. The correlation integral obeys the following scaling law:

$$\lim_{\substack{r \rightarrow 0 \\ N \rightarrow \infty}} C(r) \approx r^{D_2} \quad (14)$$

from which we can calculate D_2 as the slope of log-log curves as shown here:

$$D_2 = \lim_{r \rightarrow 0} \frac{\ln(C(r))}{\ln(r)}. \quad (15)$$

Here, $C(r)$ is a measure of the probability that two arbitrary points $\mathbf{s}(i)$ and $\mathbf{s}(j)$ of the multidimensional space will be separated by a distance r . The main point is that $C(r)$ behaves as a power of r for small values of r . Therefore, plotting $\ln(C(r))$ vs $\ln(r)$ allows us to calculate D_2 from the slopes of the curves. If the slopes of the curves for increasing embedding dimension converge to a saturation value, this limit is called the correlation dimension. For a stochastic process, there is no convergence and the slopes keep increasing with an increase in the embedding dimension.

One of the defining attributes of an attractor for which the dynamics is chaotic is that it displays exponentially sensitive dependence on initial conditions.^{18,66-71} Consider two nearby initial conditions $\mathbf{s}_1(0) = \mathbf{s}_0$ and $\mathbf{s}_2(0) = \mathbf{s}_0 + \varepsilon(0)$, and imagine that they are evolved forward in time by a continuous-time dynamical system yielding orbits $\mathbf{s}_1(t)$ and $\mathbf{s}_2(t)$. At time t , the separation between the two orbits is $\varepsilon(t) = \mathbf{s}_1(t) - \mathbf{s}_2(t)$. Suppose that for small $|\varepsilon(0)|$ and for large t , we have

$$\frac{|\varepsilon(t)|}{|\varepsilon(0)|} \approx e^{\lambda t}. \quad (16)$$

If the parameter λ , denoting the largest Lyapunov exponent, is positive, then we say that the system displays sensitive dependence on initial conditions and is chaotic.^{18,66-71} This property drastically limits the possibility of long-term prediction. Sensitivity to initial conditions leads to the fact that nearby trajectories diverge/converge exponentially with respect to time. For an n -dimensional chaotic system, there are n Lyapunov exponents (positive, zero and negative) and a measure of the divergence/convergence of its trajectories in the state-space is provided by the positive/negative Lyapunov exponents, respectively.¹⁸

Usually, the Lyapunov exponents are measured in nats/second, where nat is the natural unit of information. In our present study, we present the Lyapunov exponents in nats/sample. This is nothing but the exponents in nats/second normalized with respect to the pulse repetition frequency (PRF), that is,

$$\frac{\lambda}{\text{samples/second}} = \frac{\text{nats/second}}{\text{samples/second}} = \frac{\text{nats}}{\text{sample}}. \quad (17)$$

The Kaplan-Yorke dimension gives a meaningful measure of the dimension of a strange attractor, defined in terms of its Lyapunov exponents.²⁴⁻²⁹ In order to formulate this measure, Kaplan and Yorke examined numerically a class of systems of the form

$$x(n) = f(x(n-1)), \quad (18)$$

where $f: D \rightarrow D$ and D is a compact n -dimensional set. If there exist n Lyapunov exponents $(\lambda_1, \lambda_2, \dots, \lambda_n)$, then the dimension of the attractor is related to the Lyapunov exponents in the following way:

$$D_{KY} = K + \frac{\sum_{a=1}^K \lambda_a}{|\lambda_{K+1}|}, \quad (19)$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ and $\sum_{a=1}^K \lambda_a > 0, \sum_{a=1}^{K+1} \lambda_a < 0$. Clearly, $K < n$.

Although Kaplan and Yorke conjectured that $D_2 = D_{KY}$, in reality they need not be equal because of the fact that the estimation of D_{KY} uses only the “core” of the attractor where the trajectories of the attractor reside most of the time. (There are some discrepancies regarding whether D_{KY} is an upper bound or lower bound to D_2 as reported in the literature. Kaplan and Yorke²⁴ originally conjectured that they should be equal, but later Young^{26,27} reported that D_{KY} is a lower bound to D_2 . Later again, Grassberger and Procaccia¹⁶ suggested that D_{KY} to be an upper bound to D_2 .) The fractal dimension, in contrast, is used for measuring the point densities of the entire attractor. Thus, Lyapunov exponents provide an important link between the fractal geometry of the attractor and the property of sensitive dependence on initial conditions.

A. Phase-space reconstruction

From a signal processing perspective, an issue of paramount importance is the reconstruction of dynamics from measurements made on a single coordinate of the system. The motivation here is to make “physical sense” from the resulting time series, bypassing a detailed mathematical knowledge of the underlying dynamics. The reconstruction of dynamics using independent coordinates from a time series was first advocated by Packard *et al.*;⁷² however, this paper does not give a proof, and uses “derivative” embedding rather than time-delay embedding. The idea of time-delay or delay coordinate embedding is attributed to Ruelle and Takens. Specifically, in 1981, Takens and Mañé (independently) published mathematically profound papers on time-delay embedding, which applies to attractors that are surfaces, like a torus.^{73,74} The idea of delay coordinate mapping was refined in 1991 by Sauer *et al.*⁷⁵ The approach taken in this latter paper integrates and expands on previous results due to Whitney⁷⁶ and Takens.⁷³ The delay embedding theorem, or more precisely, the fractal delay embedding prevalence theorem, for dynamic reconstruction derived in Sauer’s article⁷⁵ holds not only in the Takens sense but also a probabilistic sense. To do so, Sauer *et al.* introduced a new theory called “prevalence” theory.

Suppose now we have a time series $s(n), n = 0, 1, \dots, N$, where N is the total number of samples and the sampling period is normalized to unity. The state-space vector $\mathbf{s}(n)$ is constructed by assigning the co-ordinates

$$\begin{aligned} s_1(n) &= s(n), \\ s_2(n) &= s(n - \tau), \\ &\vdots \\ s_{d_E}(n) &= s(n - (d_E - 1)\tau), \end{aligned} \quad (20)$$

where d_E is the *embedding dimension* and τ is the *normalized embedding delay*. These parameters are not chosen ar-

bitrarily; rather they have to be determined experimentally. Procedures for estimating the normalized embedding delay (τ) based on *Mutual Information* (MI) are described in Refs. 34, 35, and 77. For estimating the embedding dimension d_E , one may use the method of *Global False Nearest Neighbor* (GFNN) described in Refs. 9, 10, 36, and 78.

Another important parameter that needs to be determined in the analysis of an observed chaotic process is the local or dynamical dimension³⁷ (d_L). The local dimension represents the number of dynamical degrees of freedom that are active in determining the evolution of the system as it moves around the attractor.^{9,10} This integer dimension gives the number of true Lyapunov exponents of the system under investigation. It is less than or equal to the global embedding dimension; that is, $d_L \leq d_E$. The parameter d_L can be determined by using the *Local False Nearest Neighbor* (LFNN) method as described in Refs. 9, 10, and 37.

B. Estimation of correlation dimension

The correlation dimension (D_2) has been the most intensely studied invariant quantity for dynamical systems.^{3–10,16,79–82} Unlike the integer Euclidean dimension, D_2 has a fractal value.^{3–10} This dimension of a dynamical system directly reflects the way in which points on the attractor are distributed in the phase-space or embedding space.¹⁶ D_2 reflects the complexity of the underlying dynamical system, and bounds the degrees of freedom or the number of parameters required to describe the system under investigation. D_2 is estimated from the measured time series by looking at the way in which points within a sphere of radius r scales as the radius shrinks to zero in the embedding space.¹⁶ There are already a large number of very good reviews of methods for the estimation of D_2 in the literature.^{9,10,16,29,30,33,79–94} The algorithm developed by Grassberger and Procaccia (GPA) is perhaps the most widely used one for computing D_2 .¹⁶ Basically, GPA measures the rate of change of the local point densities around the attractor surface as a function of the local radius. However, this algorithm suffers from certain disadvantages: (1) highly demanding computational requirements,^{30,95,96} and (2) spurious estimates of D_2 due to the presence of measurement noise. For a data length of N , GPA needs a computation on the order of $O(N^2)$ for one local center and one embedding dimension. This has to be repeated for more and more embedding dimensions. Moreover, a reliable estimate of D_2 using GPA requires a large amount of noise-free data. GPA fares poorly in the presence of noise (for SNR less than 20 dB), where the plot of $\ln(C(r)) - \ln(r)$ may not even converge. Further, the scaling region for determining the slope (and hence D_2) is more often arbitrarily fixed, thereby giving rise to varying results by different investigators.⁹²

These limitations of GPA motivated us to use a more reliable method, namely, one based on the maximum likelihood principle for the estimation of D_2 .^{84,85} Maximum likelihood is regarded as a reliable method for parameter estimation. Schouten *et al.* have developed such a method for estimating D_2 in the presence of noise.³³ We refer to this

algorithm as the STB algorithm in recognition of its originators, Schouten, Takens and Bleek.

C. Estimation of Lyapunov spectrum

The correlation dimension (D_2) characterizes the distribution of points in the state space, whereas Lyapunov exponents describe the action of the dynamics defining the evolution of the trajectories.^{18,66–71} They are the average exponential rates of divergence/convergence of nearby trajectories in phase-space.¹⁸ The Lyapunov exponents of a dynamical system are the most important invariants that characterize the system. They are independent of initial conditions on any orbit and are thus properties of the attractor geometry and the dynamics.^{9,18,66–71} For a system to be chaotic, it has to have at least one positive Lyapunov exponent. The magnitude of this exponent defines the time scale on which the system dynamics become unpredictable.⁹⁷ Bennettin *et al.* showed that the estimation of Lyapunov exponents for systems, whose equations of motion are well defined, is straightforward.^{69,70,98} Unfortunately, their method cannot be applied directly to experimental time series. Wolf's method is only suitable for estimating the largest positive Lyapunov exponent of a chaotic process.¹⁸ Moreover, it requires data with high SNR and the attractor dimension should be medium or low; these are difficult conditions to obtain in an experimental set-up.

There is an abundance of literature available on different methods for the estimation of Lyapunov exponents using an experimental time series.^{9,10,18,26–29,38–40,66–71,99–105} A frequently used method for estimating the Lyapunov exponents of an experimental time series involves recursive QR decomposition applied to the Jacobian of a function, $\mathbf{F}[\mathbf{s}(n)]$, which maps points on the orbit into points at t time steps later:^{9,10,38–40}

$$\mathbf{s}(n+t) = \mathbf{F}[\mathbf{s}(n)]. \quad (21)$$

Here the map is unknown, and the Jacobian is to be estimated using knowledge of the trajectory points $\mathbf{s}(n)$. The map $\mathbf{F}$ describes nearby trajectories as well as how the distance between these neighbors changes with time. Let $\mathbf{z}^{(r)} \times (n, 0)$ be the distance between the fiducial orbit and the r th neighbor at $t=0$. The corresponding distance, time t later, is

$$\mathbf{z}^{(r)}(n, t) = \mathbf{F}[\mathbf{s}(n) + \mathbf{z}^{(r)}(n, 0)] - \mathbf{F}[\mathbf{s}(n)]. \quad (22)$$

A Taylor series expansion of $\mathbf{F}[\cdot]$ contains the Jacobian of the underlying dynamics as the leading term of this expression. The terms in the Jacobian are found by a least-squares minimization of the residuals in the above formula.^{38–40} The Lyapunov exponents are then calculated as the logarithm of the eigenvalues of products of such Jacobian matrices for a number of steps along the trajectory of the orbit.^{10,106} We refer to this algorithm as the BBA algorithm, in recognition of its originators, Brown, Bryant and Abarbanel.^{38,39} In an independent publication, Briggs reported an algorithm similar to that of Brown *et al.* in the same year.⁴⁰

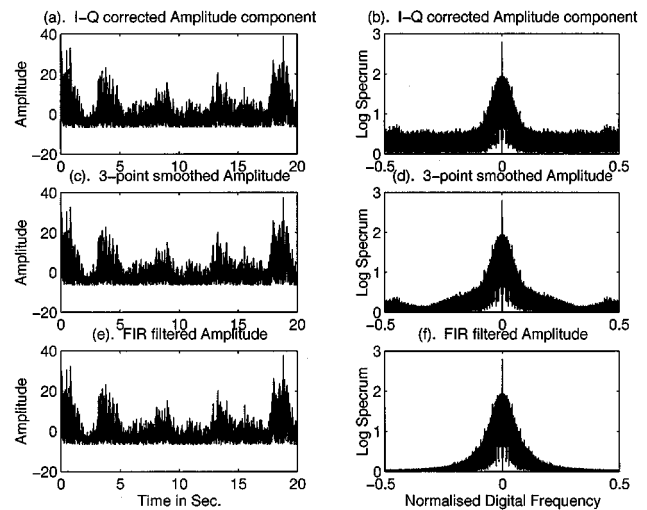


FIG. 6. Figure showing the effect of filtering: (a) I-Q corrected amplitude component of sea clutter, (b) the corresponding FFT spectrum, (c) 3-point smoothed amplitude, and (d) its spectrum, (e) FIR filtered amplitude, and (f) its spectrum.

IX. RESULTS

In this section, we discuss the results of our experimental studies on the five sets of sea clutter data parametrized in Section III. The first subsection discusses the I-Q calibration, filtering, testing for continuity and differentiability, stationarity, and nonlinearity. In the second subsection, results on the D_2 analysis are presented. In the third and fourth subsections, the results on Lyapunov exponents and Lyapunov dimension estimation are presented, respectively.

A. Preprocessing

Each sea clutter data set had at least 50 000 samples taken from a fixed resolution cell on successive sweeps. The sampling period is determined by the pulse repetition interval of the radar. In each case, the data were preprocessed for amplitude and phase correction (coherent radar data), as described in Section V. Once the I-Q calibration was done, the data (I and Q components) were subjected to filtering (3-point smoothing or FIR filtering) before computing the amplitude (in the case of coherent radar data). In the case of noncoherent radar data sets, amplitude of the clutter data is available in the first place, and the filtering is performed on the amplitude data alone. Figure 6 illustrates the unfiltered and filtered amplitude components of sea clutter data for 20 second durations; Fig. 6(a) refers to unfiltered data, and Figs. 6(c) and 6(e) refer to filtered data. The corresponding FFT spectra are shown in Figs. 6(b), 6(d) and 6(f), respectively. We observe that the unfiltered amplitude signal has high-frequency noise components, as its FFT spectrum clearly indicates. Figure 6(d), which is the FFT spectrum of the 3-point smoothed amplitude component, clearly shows the attenuation of the high-frequency components without affecting the actual clutter spectrum. The attenuation of noise is almost complete in the second filtered case shown in Fig. 6(f), where we used a 100-tap FIR filter for suppressing the

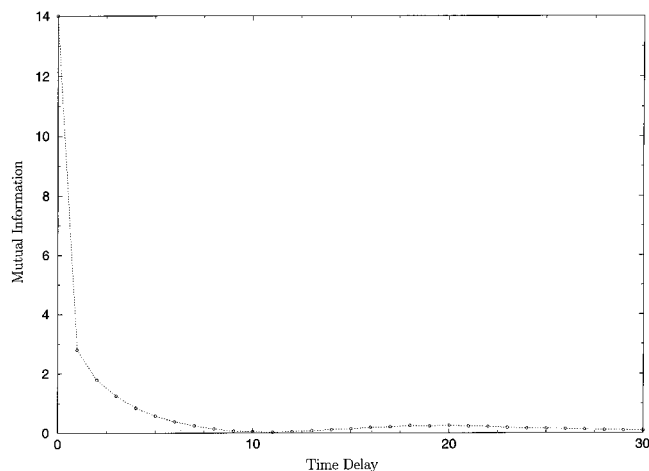


FIG. 7. The MI for a typical sea clutter data set.

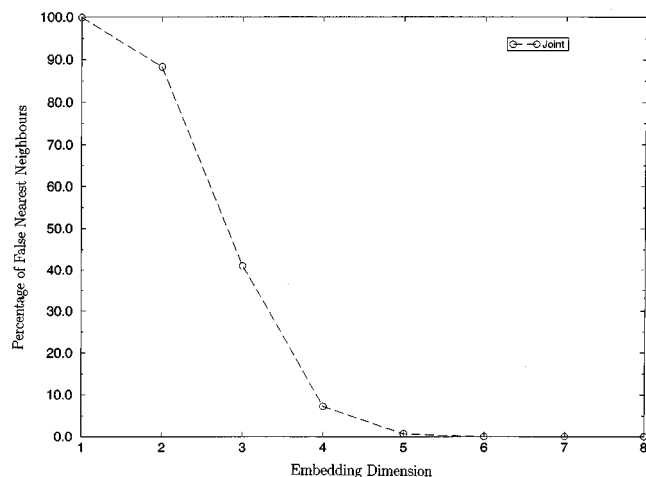


FIG. 8. The GFNN for a typical sea clutter data set.

high-frequency noise from the amplitude component. Here again, the filtering did not alter the spectrum of the sea clutter. After these preprocessing procedures were completed, each data set was analyzed to test whether the preprocessing had altered the underlying dynamics.^{57–62} This was done by statistically testing for the continuity and differentiability between the original and preprocessed data embedding, as described in Section VI.

The determination of embedding parameters (i.e., the embedding dimension d_E and normalized embedding delay τ) is an essential step before performing the statistical tests. These parameters were obtained by using the GFNN and MI algorithms, respectively.^{9,10,34–36,38,78} Figure 7 shows the results obtained using the MI algorithm for one particular clutter data set. Here we used an embedding dimension $d_E=2$ and the normalized embedding delay varied from 0 to 50. The optimum embedding delay τ happened to be 11, where the MI reaches a minimum for the first time. Note that the corresponding minimum value of the mutual information is small, which indicates that radar samples spaced by $\tau=11$ time units apart are essentially independent; and yet it is non-zero, which indicates that the radar samples so delayed are correlated with each other. The optimum embedding dimension for the same data was found to be $d_E=5$ from the GFNN analysis. Figure 8 shows the percentage (%) GFNN versus different embedding dimensions. We used thresholds R_{tol1} and R_{tol2} of 25 and 10, respectively, and a normalized embedding delay $\tau=11$ (obtained from the MI analysis) for this calculation. This procedure was repeated for the entire database used for the study.

The LFNN analysis for estimating the local embedding dimension (d_L) of sea clutter data were performed for about 30 data sets drawn from different categories of the data sets used for the study. The analysis included data sets from different wave heights of sea surface, raw (I-Q corrected), 3-point smoothed and FIR filtered in-phase, quadrature-phase and amplitude components from different radar sites. Results of this study show that the local dimension varies between 5 and 6. Figure 9 illustrates the results of the local false nearest

neighbor analysis on the same clutter data used in Fig. 8. The four curves shown in Fig. 9 correspond to local nearest neighbors (N_B) of 40, 60, 80, and 100. It may be observed that beyond an embedding dimension $d_L=5$, the percentage of bad predictions is almost independent of both the choice of d_L and the choice of N_B . Figure 9 implies that a local dimension $d_L=5$ is adequate for the analysis; this result defines the number of true Lyapunov exponents. Note also that for the same data set used in Fig. 8, we have $d_E=5$, which is in agreement with the requirement that $d_E \geq d_L$.

Figure 10 shows the continuity statistic⁶³ ($\Theta_{C0}(\epsilon)$) for one typical clutter (amplitude before and after filtering) time series of various lengths (1000, 10 000, and 50 000). We have chosen an embedding dimension $d_E=5$, embedding delay $\tau=11$, number of randomly chosen centers = 50, and ϵ values ranging from 0.001 to 1.0. As expected, the continuity drops when the ϵ value becomes smaller and smaller. This results from a simple lack of points in the small sets, so the null hypothesis cannot be rejected for few random centers around the attractor. However, when more points are added to the time series, confidence in continuity goes up at smaller

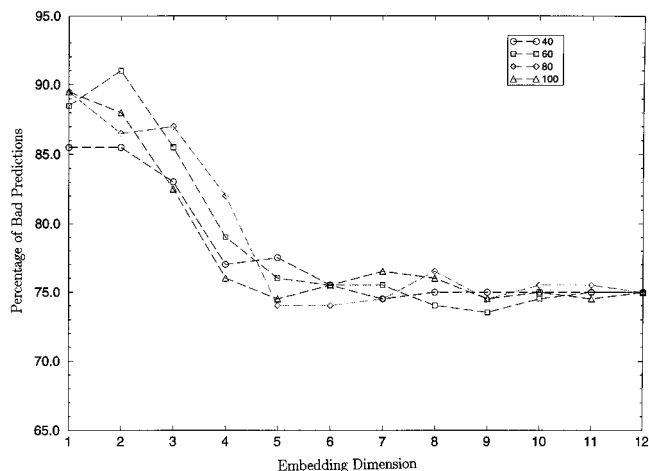
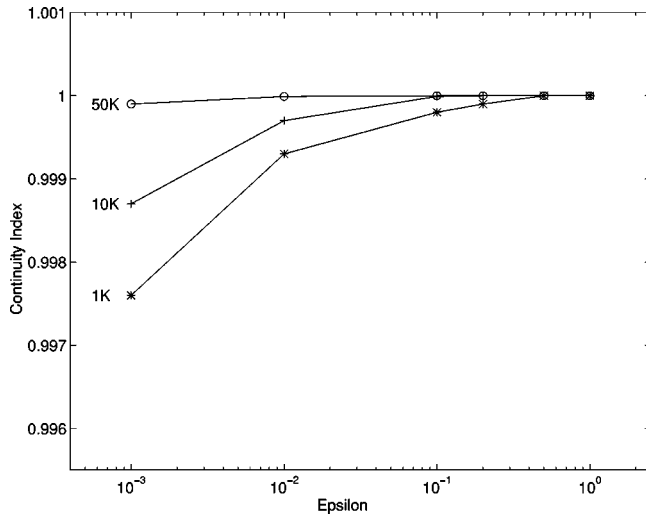


FIG. 9. The LFNN for the same sea clutter data used in Fig. 8.

FIG. 10. The continuity index $\Theta_{C^0}(\varepsilon)$ for sea clutter data.

ϵ values also. When the mapping is continuous, the respective statistic continually improves and approaches 1.0 asymptotically. The continuity statistic $\Theta_{C^0}(\varepsilon)$ varies between 0.99 and 1.0. This is what we observe in Fig. 10.

Figure 11 illustrates the differentiability statistic⁶³ ($\Theta_{C^1}(\varepsilon)$) for the same data set. Here also, the differentiability statistic ($\Theta_{C^1}(\varepsilon)$) approaches 1.0 asymptotically. The $\Theta_{C^1}(\varepsilon)$ varies between 0.93 and 1.0 (see Fig. 11). We extended these tests for several such clutter data sets, and a similar performance was observed. In all the cases, the filtered data passed the test, meaning that the underlying dynamics of experimental sea clutter data remained unchanged after the preprocessing.

After completing these tests, the data were subjected to tests for stationarity, using recurrence plots.⁶⁴ Figure 12 shows the recurrence plot for a typical sea clutter data set. Here, we used 30 000 data points with an embedding dimension $d_E=5$, normalized embedding delay $\tau=11$ and a num-

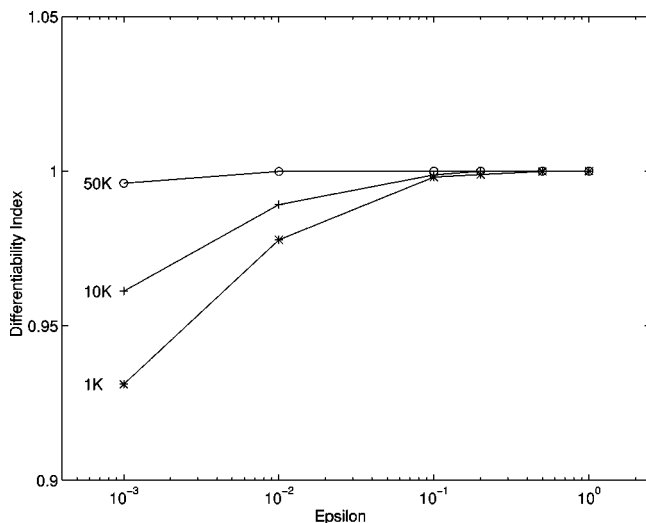
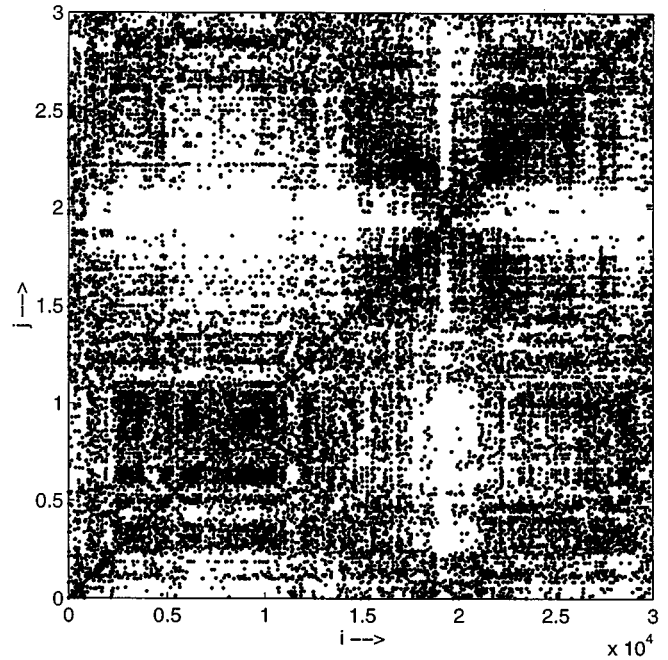
FIG. 11. The differentiability index $\Theta_{C^1}(\varepsilon)$ for sea clutter data.

FIG. 12. A recurrence plot for sea clutter data.

ber of nearest neighbors equal to 10. The overall pattern is fairly uniform, confirming the stationarity of the data set.

Next, we discuss some of the results we obtained from the nonlinearity tests on clutter data sets. As we mentioned previously, the nonlinearity tests were performed in three different ways.

In the first (WSF) method, we designed a wave-shaping filter to filter out a pure Gaussian white sequence (linear stochastic process generated by a commercially available physical device) to produce a signal that has the same power spectrum as that of the clutter data set. We used a data length of 4096 points of the original sea clutter data for designing the wave-shaping filter. Figure 13 shows the experimental data details. Figure 13(a) refers to the original sea clutter data, and its power spectrum is shown in Fig. 13(b). Figure 13(e) is the time series obtained using the wave-shaping filter stimulated by white Gaussian noise [Fig. 13(c)]. Its spectrum is shown in Fig. 13(f), which is almost the same as that in Fig. 13(b) referring to the original sea clutter data set.

The set of criteria listed in Section IV is applied to the signals shown in Fig. 13(a) (original sea clutter data) and Fig. 13(e) (filtered signal using the wave shaping filter), independently, to determine the nature of the underlying physical process by which the signals are being generated. The filtered signal behaved like a noise process, whereas the sea clutter data revealed that it is indeed a chaotic process. The results of these studies on these two signals are listed below:

(a). *Filtered signal*

- (1) The nonlinearity test using the SIPD algorithm showed that the filtered signal is a linear process (known fact) with a Mann-Whitney rank-sum statistic⁶⁵ (Z-value) always positive (>0).
- (2) The correlation dimension D_2 was fractal but it kept on

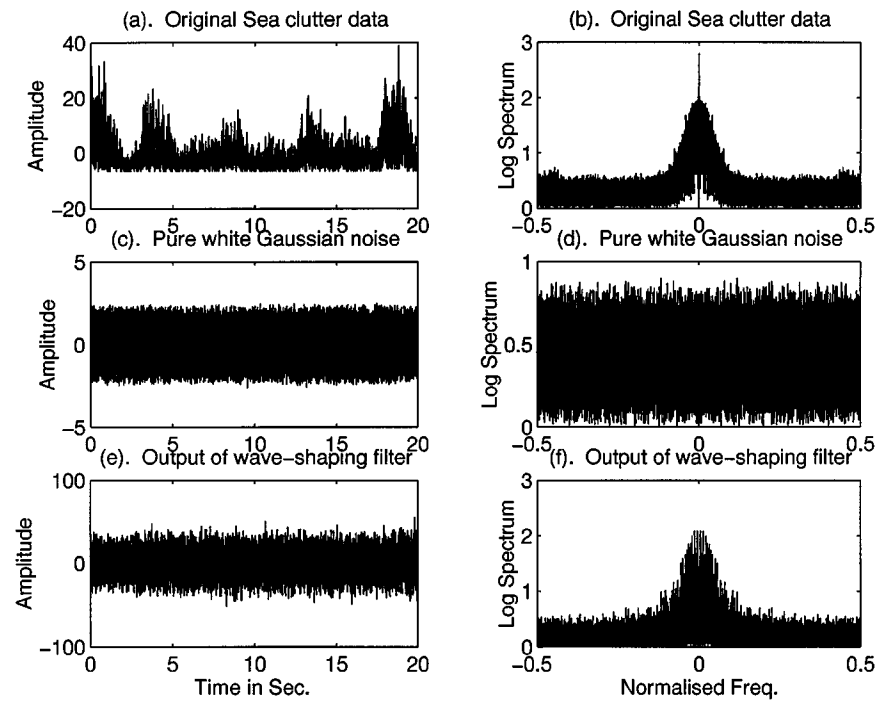


FIG. 13. Figure showing the WSF procedure: (a) I-Q corrected amplitude component of sea clutter, (b) the corresponding FFT spectrum, (c) pure white Gaussian noise, and (d) its spectrum, and (e) output of the WSF, and (f) its spectrum.

increasing for increasing embedding dimension (d_E) values. This is again a characteristic of a noise process.

- (3) The Lyapunov spectrum consisted of several positive exponents and the number of positive exponents kept on increasing for increasing values of d_E . The sum of all Lyapunov exponents for $d_E < 7$ was always positive, showing the non-dissipative nature of the system.
- (4) Computation of the Kaplan-Yorke dimension (D_{KY}) was not feasible for $d_E < 7$. For higher d_E 's, the computed D_{KY} and the estimated D_2 were never close.

(b). Actual sea clutter data

- (1) The data passed the nonlinearity test with a Z-value⁶⁵ always less than -3.0 .
- (2) The D_2 value was fractal and it saturated to a value between 4.1 to 4.5 for increasing embedding dimension.
- (3) The Lyapunov spectrum consisted of two positive exponents followed by one exponent close to zero and two or more negative exponents corresponding to embedding dimension 5 or more (the number of exponents is equal to the embedding dimension). The sum of all Lyapunov exponents was always negative, indicating that it is indeed a dissipative system.
- (4) The computed D_{KY} was always close to the estimated D_2 .

The results of this first method show that sea clutter is a nonlinear process.

Next, we applied the second (SIPD) method to test the nonlinearity of the sea clutter data sets.³¹ The Mann-Whitney rank-sum statistics (Z-value) for 50 surrogate data sets were less than -3.0 for almost all the data sets used.⁶⁵ Therefore,

the test rejected the null hypothesis that each clutter sample is that of a linear process.

In the third (SCD) method, the surrogate data analysis proposed by Theiler *et al.* was applied to a minimum of 10 sea clutter data sets.³² Twenty such phase-shuffled surrogate sets were generated for each clutter data set. The surrogate D_2 (maximum likelihood estimate) values were statistically compared (Student's *t* test) with the D_2 values of the original clutter data.⁶⁵ The surrogate sets were significantly different from the original data sets ($p < 0.0001$). Figure 14 shows the original data [Fig. 14(a)] and their two surrogates [Figs. 14(c) and 14(e)] along with their Fourier spectra (Figs. 14(b), (d) and (f), respectively). The waveform of the surrogate data looks entirely different from that of the original sea clutter signal, but their spectra are essentially identical. The spurious low-frequency effects observed in Figs. 14(b), 14(d) and 14(f) are due to the limitation of the algorithm.³² From this nonlinearity test also, the inference is that sea clutter is a nonlinear process.

Thus these three nonlinearity tests, in their own individual ways, confirm the known fact that the underlying physics of sea clutter is indeed nonlinear.

B. Correlation dimension

As mentioned earlier, the correlation dimension D_2 is the most basic and static property of an attractor. We used the STB algorithm for estimating the D_2 of sea clutter data.^{31,33}

In this work, we made an extensive study of D_2 of sea clutter, using the five data sets parametrized in Section III. These data sets were collected using different radar sites,

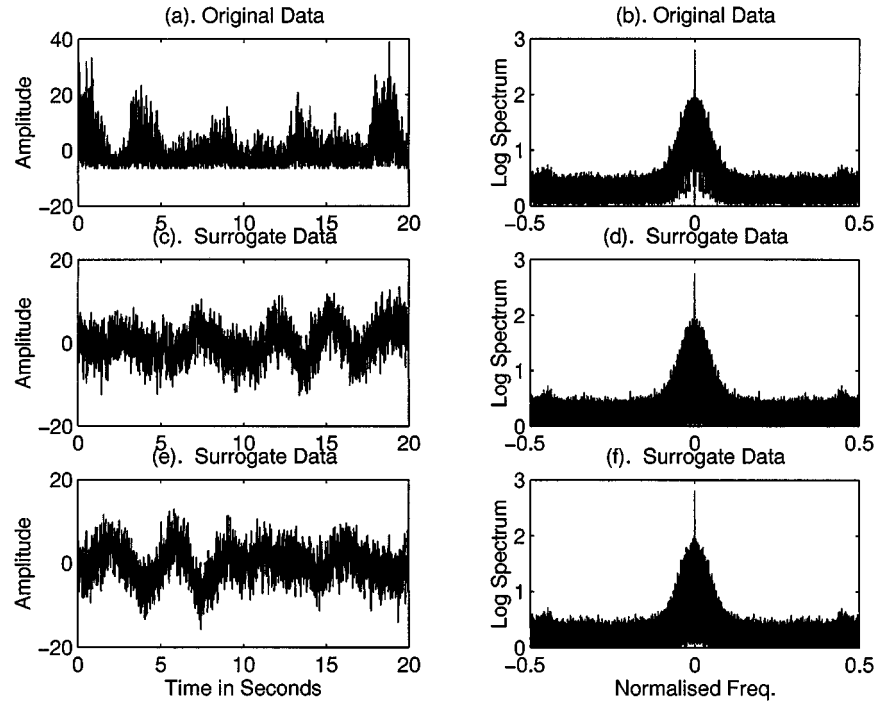


FIG. 14. Figure showing the surrogate data analysis: (a) I-Q corrected amplitude component of sea clutter, (b) the corresponding FFT spectrum, (c) one of the surrogate data, (d) its spectrum, (e) another surrogate data, and (f) its spectrum.

different radar parameters, and different sea states. (*Sea state* is a single number that provides a gross summary of the degree of agitation of the sea and the characteristics of its surface; sea state and wave height are related in some fashion.¹⁰⁷) Each data set was 50 000 samples long. We thoroughly investigated the effects of the following:

- (i) radar component: in-phase, quadrature-phase, and amplitude,
- (ii) pulse duration,
- (iii) radar range,
- (iv) pulse repetition frequency,
- (v) like-polarization (HH or VV), and
- (vi) sea state.

In each case, the following preprocessings were performed before extracting the chaotic parameters:

- (i) I-Q calibration,
- (ii) 3-point smoothing, and
- (iii) FIR filtering.

The embedding parameters d_E and τ of the sea clutter data were obtained using the GFNN and MI algorithms, respectively.¹⁰⁶ The D_2 values for different data sets obtained using the STB algorithm show that the value varies between 4.1 and 4.5, irrespective of changes in the environmental conditions and radar parameters. It gives slightly higher estimates of D_2 in the case of unfiltered data where the signal-to-noise ratio is low, which is to be expected because of the presence of noise. Apart from this fact, the estimated values were consistent for wide variations in sea clutter data. Tables I–III show the D_2 values for different

ranges of I-Q corrected, 3-point smoothed and FIR filtered in-phase, quadrature-phase, and amplitude components of clutter data using the Dartmouth database. The maximum likelihood estimate of D_2 and Kaplan-Yorke dimension are shown side-by-side for each data set. The pulse duration values for these three tables are 2000, 200 and 200 ns, respectively. We observe that the pulse duration has essentially no

TABLE I. Correlation dimension of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set I, Dartmouth).

Range (m)	Correlation dimension					
	I-Q corrected		3-point smoothed		FIR filtered	
	D_{KY}	D_{ML}	D_{KY}	D_{ML}	D_{KY}	D_{ML}
Amplitude						
1800	4.59	4.24	4.30	4.40	4.35	4.42
2100	4.34	4.37	4.28	4.41	4.43	4.32
2400	4.60	4.42	4.30	4.52	4.48	4.45
2700	4.34	4.36	4.31	4.43	4.38	4.53
3000	4.48	4.36	4.29	4.43	4.49	4.50
In-phase component						
1800	4.25	4.20	4.23	4.22	4.25	4.16
2100	4.26	4.30	4.31	4.24	4.35	4.20
2400	4.27	4.20	4.32	4.25	4.38	4.20
2700	4.15	4.18	4.29	4.01	4.34	4.01
3000	5.02	4.21	4.27	4.10	4.30	4.10
Quadrature-phase component						
1800	4.27	4.42	4.27	4.10	4.27	4.07
2100	4.25	4.21	4.34	4.24	4.25	4.28
2400	4.24	4.27	4.37	4.20	4.37	4.20
2700	5.12	4.16	4.35	4.10	4.34	4.04
3000	5.09	4.26	4.30	4.23	4.27	4.12

TABLE II. Correlation dimension of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set II, Dartmouth).

Range (m)	Correlation dimension					
	I-Q corrected		3-point smoothed		FIR filtered	
	D_{KY}	D_{ML}	D_{KY}	D_{ML}	D_{KY}	D_{ML}
Amplitude						
1590	4.27	4.28	4.18	4.70	4.47	4.33
1620	4.20	4.57	4.20	4.31	4.44	4.78
1650	4.31	4.41	4.10	4.33	4.37	4.51
1680	4.69	4.28	4.45	4.53	4.36	4.69
1710	4.33	4.35	4.47	4.46	4.30	4.25
In-phase component						
1590	4.87	4.50	4.43	4.28	4.64	4.61
1620	4.85	4.42	4.20	4.35	4.41	4.21
1650	4.10	4.10	4.21	4.13	4.31	4.05
1680	4.12	4.75	4.17	4.36	4.46	4.17
1710	4.40	4.50	4.18	4.17	4.41	4.11
Quadrature-phase component						
1590	4.32	4.75	4.23	4.16	4.45	4.38
1620	4.13	4.85	4.19	4.07	4.46	4.43
1650	4.45	4.91	4.40	4.53	4.44	4.05
1680	4.10	4.27	4.54	4.68	4.35	4.19
1710	4.02	4.23	4.31	4.81	4.36	4.45

effect on the D_2 values. Similarly, for different starting range values (1800, 1590, and 2739 m), the D_2 values did not show any dependence on range.

The effect of like-polarization of the radar, HH or VV, was also evaluated. Here too we see that the D_2 value is essentially independent of like-polarization. Table I data were obtained using horizontal like-polarization, and Tables II and III data were obtained using vertical like-polarization. These three tables pertain to the same radar type and the same radar site.

TABLE III. Correlation dimension of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set III, Dartmouth).

Range (m)	Correlation dimension					
	I-Q corrected		3-point smoothed		FIR filtered	
	D_{KY}	D_{ML}	D_{KY}	D_{ML}	D_{KY}	D_{ML}
Amplitude						
2739	4.93	4.74	4.20	4.39	4.27	4.67
2799	4.23	4.17	4.22	4.38	4.28	4.11
2859	4.30	4.38	4.24	4.28	4.29	4.46
2919	4.36	4.24	4.28	4.26	4.33	4.24
2979	4.35	4.26	4.37	4.44	4.35	4.21
In-phase component						
2739	4.62	4.28	4.41	4.20	4.59	4.17
2799	4.71	4.79	4.44	4.14	4.69	4.31
2859	4.72	4.24	4.86	4.60	4.72	4.30
2919	4.68	4.40	4.52	4.55	5.70	4.40
2979	4.97	4.21	4.52	4.36	4.95	4.16
Quadrature-phase component						
2739	4.62	4.54	4.37	4.38	4.60	4.31
2799	4.77	4.50	4.48	4.52	4.26	4.65
2859	4.71	4.26	4.89	4.39	4.72	4.38
2919	5.00	4.58	4.46	4.24	5.70	4.36
2979	4.96	4.42	4.43	4.60	4.96	4.25

TABLE IV. Correlation dimension of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set IV, Cape Bonavista).

Range (m)	Correlation dimension					
	I-Q corrected		3-point smoothed		FIR filtered	
	D_{KY}	D_{ML}	D_{KY}	D_{ML}	D_{KY}	D_{ML}
Amplitude						
5730	4.86	4.25	4.11	4.17	4.17	4.37
5850	5.78	4.46	4.25	4.19	4.21	4.09
6000	4.80	4.22	4.13	4.10	4.16	4.02
6150	4.93	4.57	4.20	4.31	4.21	4.04
6300	4.99	4.61	4.17	4.21	4.14	4.21
6450	4.57	4.54	4.15	4.26	4.24	4.24
6600	4.31	4.63	4.12	4.17	4.15	4.24
In-phase component						
5730	4.46	4.69	4.16	4.32	4.23	4.10
5850	4.81	4.70	4.10	4.32	4.16	4.19
6000	4.55	4.27	4.20	4.29	4.30	4.03
6150	4.45	4.84	4.32	4.23	4.34	4.11
6300	4.70	4.48	4.30	4.14	4.22	4.11
6450	4.39	4.97	4.10	4.32	4.13	4.06
6600	4.20	4.52	4.11	4.19	4.05	4.15
Quadrature-phase component						
5730	4.99	4.25	4.10	4.07	4.23	4.34
5850	5.07	4.30	4.13	4.33	4.18	4.17
6000	4.60	4.40	4.26	4.28	4.38	4.10
6150	5.30	4.57	4.52	4.63	4.51	4.10
6300	4.50	4.61	4.21	4.38	4.32	4.14
6450	4.35	4.53	4.10	4.06	4.14	4.14
6600	4.75	4.62	4.10	3.95	4.16	4.08

Moreover, we observe that Tables I–V (which correspond to different radar locations) are more or less consistent in terms of the D_2 values, irrespective of the fact that the data were recorded at different radar locations and using different radar types.

The effect of varying sea state was also studied. The sea states, measured in terms of wave height, change from 0.85 m to 2.6 m in Tables I–V. We see that the D_2 values are essentially independent of wave height and therefore sea state. Table V shows the D_2 estimates of raw, 3-point smoothed, and FIR filtered clutter data from the experiments at Argentina. Here, it was noted that the raw data were more noisy (low SNR), and because of the influence of noise, the D_2 estimates show higher values for the unfiltered dataset. However, the D_2 estimates of the filtered data once again are in the range of 4.1 to 4.5.

TABLE V. Correlation dimension of amplitude of clutter data for different ranges (data set V, Argentina).

Range (m)	Correlation dimension					
	Raw		3-point smoothed		FIR filtered	
	D_{KY}	D_{ML}	D_{KY}	D_{ML}	D_{KY}	D_{ML}
4500	5.63	5.62	4.26	4.41	4.14	4.32
5625	...	7.94	4.96	5.20	4.41	4.79
6900	6.70	6.92	4.38	4.69	4.27	4.37

TABLE VI. Lyapunov exponents of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set I, Dartmouth).

Range (m)	Lyapunov exponents (nats/sample)								
	Amplitude			In-phase component			Quadrature-phase component		
	Corrected	3 point	FIR	Corrected	3 point	FIR	Corrected	3 point	FIR
1800	+0.1927	+0.2069	+0.2047	+0.1333	+0.1168	+0.0969	+0.1481	+0.1195	+0.0948
	+0.0966	+0.0947	+0.0876	+0.0558	+0.0387	+0.0244	+0.0539	+0.0401	+0.0312
	+0.0047	-0.0128	+0.0022	-0.0138	-0.0042	-0.0040	-0.0099	-0.0087	-0.0070
	-0.1157	-0.1461	-0.1289	-0.0909	-0.0841	-0.0585	-0.1018	-0.0750	-0.0525
	-0.3023	-0.4825	-0.4753	-0.3602	-0.2954	-0.2383	-0.3443	-0.2799	-0.2438
2100	-0.6793								
	+0.1554	+0.2044	+0.2076	+0.1155	+0.1191	+0.0927	+0.1062	+0.1128	+0.0861
	+0.0766	+0.0835	+0.0973	+0.0476	+0.0450	+0.0310	+0.0508	+0.0429	+0.0246
	-0.0132	-0.0126	+0.0020	-0.0134	-0.0089	-0.0001	-0.0055	-0.0024	-0.0082
	-0.1267	-0.1412	-0.1196	-0.0802	-0.0730	-0.0531	-0.0870	-0.0672	-0.0460
2400	-0.2763	-0.4712	-0.4323	-0.2757	-0.2678	-0.2039	-0.2663	-0.2533	-0.2219
	-0.6840								
	+0.1951	+0.2246	+0.2519	+0.1141	+0.1304	+0.0931	+0.1152	+0.1182	+0.0981
	+0.1084	+0.1137	+0.1146	+0.0450	+0.0496	+0.0260	+0.0402	+0.0444	+0.0294
	+0.0009	-0.0044	+0.0054	-0.0087	-0.0107	-0.0016	-0.0100	-0.0072	-0.0052
2700	-0.1189	-0.1734	-0.1378	-0.0804	-0.0764	-0.0428	-0.0763	-0.0657	-0.0458
	-0.3082	-0.5295	-0.4885	-0.2737	-0.2907	-0.1978	-0.2932	-0.2458	-0.2043
	-0.7537								
	+0.1619	+0.1968	+0.2093	+0.0967	+0.1040	+0.0790	+0.1165	+0.1043	+0.0937
	+0.0668	+0.0948	+0.0993	+0.0361	+0.0417	+0.0296	+0.0539	+0.0421	+0.0271
3000	-0.0119	-0.0056	+0.0036	-0.0127	-0.0052	-0.0069	+0.0047	+0.0025	-0.0012
	-0.1140	-0.1418	-0.1427	-0.0823	-0.0718	-0.0392	-0.0325	-0.0667	-0.0542
	-0.3045	-0.4605	-0.4457	-0.2648	-0.2378	-0.1849	-0.1055	-0.2335	-0.1955
	-0.7151						-0.3126		
	+0.1839	+0.2198	+0.2119	+0.0939	+0.0897	+0.0949	+0.1147	+0.0964	+0.0820
	+0.0822	+0.1028	+0.1080	+0.0378	+0.0331	+0.0231	+0.0536	+0.0357	+0.0307
	-0.0023	-0.0180	+0.0016	+0.0077	-0.0097	-0.0089	+0.0075	-0.0023	-0.0068
	-0.1171	-0.1596	-0.1046	-0.0277	-0.0552	-0.0471	-0.0369	-0.0621	-0.0515
	-0.3037	-0.5063	-0.4378	-0.1057	-0.2149	-0.2120	-0.1091	-0.2246	-0.2039
	-0.7746			-0.3000			-0.3095		

These results show that sea clutter has a strange attractor with a correlation dimension lying between 4.1 to 4.5. This result holds irrespective of radar type, radar location, or sea state. We also observe that the D_2 values for in-phase, quadrature-phase and amplitude components remain essentially consistent for all data sets. This latter observation indicates that a single radar component should be sufficient for reconstructing the underlying dynamics of the sea clutter in accordance with Takens embedding theorem.⁷²⁻⁷⁴

C. Lyapunov spectrum

As mentioned previously, the Lyapunov exponents tell us not only about the sensitive dependence of dynamics on the initial conditions of nearby trajectories, but also how orbits on the attractor move apart/together under the evolution of the underlying dynamics.

We used the BBA algorithm for estimating the Lyapunov exponents of the sea clutter data.^{9,10,38,39} The algorithm gave 5 or 6 exponents corresponding to the embedding dimension used. A data length of 50 000 samples and a minimum of 3000 starting locations were used in each calculation. Usually, the Lyapunov exponents are measured in

units of nats per second. However, in our experimental study, the sampling period was normalized with respect to the pulse repetition frequency of the radar. Accordingly, the Lyapunov exponents values reported in this paper are measured in units of nats per sample as explained in Eqn. (17).

Here again, as we did in the D_2 analysis, the effects of varying radar parameters and sea states on the Lyapunov spectrum were studied in detail. Tables VI–X summarize the Lyapunov spectra for the five sea clutter data sets parametrized in Section III. From these tables, we observe the following:

- (i) For a given radar type and sea state, the Lyapunov exponents are essentially independent of radar range, and radar component (in-phase, quadrature-phase, or amplitude).
- (ii) For a given radar type, the Lyapunov spectrum exhibits significant dependence on sea state (see Tables VII and VIII pertaining to the same radar system and radar site but different wave heights). In particular, the magnitudes of the positive Lyapunov exponents in-

TABLE VII. Lyapunov exponents of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set II, Dartmouth).

Range (m)	Lyapunov exponents (nats/sample)								
	Amplitude			In-phase component			Quadrature-phase component		
	Corrected	3 point	FIR	Corrected	3 point	FIR	Corrected	3 point	FIR
1590	+0.1998	+0.2325	+0.2693	+0.2730	+0.2692	+0.2929	+0.1888	+0.1728	+0.2422
	+0.0938	+0.1060	+0.1458	+0.1557	+0.1276	+0.1423	+0.0883	+0.0646	+0.1100
	-0.0207	-0.0260	+0.0183	+0.0353	+0.0025	+0.0208	-0.0111	-0.0089	+0.0018
	-0.1658	-0.2017	-0.1603	-0.1248	-0.1537	-0.1292	-0.1445	-0.1237	-0.1193
	-0.3994	-0.6106	-0.5735	-0.3897	-0.5713	-0.5070	-0.3861	-0.4626	-0.5154
1620	-0.9518			-1.0103			-0.9760		
	+0.1976	+0.2334	+0.2804	+0.2890	+0.1855	+0.1983	+0.1671	+0.1666	+0.2158
	+0.0916	+0.1064	+0.1416	+0.1663	+0.0759	+0.0810	+0.0615	+0.0584	+0.1006
	-0.0229	-0.0278	+0.0096	+0.0388	-0.0098	-0.0012	-0.0280	-0.0120	-0.0003
	-0.1889	-0.2062	-0.1630	-0.1444	-0.1482	-0.0978	-0.1502	-0.1262	-0.1132
1650	-0.4247	-0.6265	-0.6012	-0.4100	-0.5094	-0.4429	-0.3870	-0.4469	-0.4399
	-0.9928			-1.0410			-0.9729		
	+0.2151	+0.2068	+0.2482	+0.2242	+0.1822	+0.1434	+0.1314	+0.1231	+0.2286
	+0.0980	+0.0871	+0.1229	+0.1115	+0.0828	+0.0616	+0.0432	+0.0478	+0.1139
	-0.0190	-0.0477	-0.0005	-0.0293	-0.0124	-0.0049	-0.0425	-0.0126	+0.0117
1680	-0.1600	-0.2112	-0.1666	-0.2570	-0.1460	-0.0908	-0.1660	-0.0849	-0.1295
	-0.4318	-0.6378	-0.5537	-0.7535	-0.5029	-0.3567	-0.3679	-0.1853	-0.5037
	-1.0059						-0.9129	-0.5206	
	+0.2160	+0.1972	+0.2544	+0.1421	+0.1662	+0.1862	+0.1590	+0.1268	+0.1672
	+0.1129	+0.1048	+0.1263	+0.0588	+0.0622	+0.0893	+0.0567	+0.0575	+0.0727
1710	+0.0021	-0.0093	+0.0007	-0.0248	-0.0166	-0.0037	-0.0308	-0.0072	-0.0092
	-0.1261	-0.1398	-0.1723	-0.1340	-0.1373	-0.1010	-0.1659	-0.0799	-0.1035
	-0.2974	-0.3404	-0.5780	-0.3446	-0.4312	-0.3700	-0.3746	-0.1768	-0.3647
	-1.1682	-0.7734		-0.7468			-0.8638	-0.4654	
	+0.2051	+0.1873	+0.2252	+0.1828	+0.1746	+0.2068	+0.1403	+0.0983	+0.2206
	+0.1050	+0.1019	+0.1135	+0.0900	+0.0683	+0.0918	+0.0485	+0.0356	+0.1051
	-0.0073	-0.0072	-0.0125	-0.0066	-0.0186	-0.0015	-0.0300	-0.0132	-0.0012
	-0.1606	-0.1284	-0.1640	-0.1287	-0.1410	-0.1151	-0.1523	-0.0709	-0.1516
	-0.4342	-0.3246	-0.5337	-0.3486	-0.4691	-0.4366	-0.3457	-0.1612	-0.4900
	-1.0249	-0.7560		-0.8480			-0.7810	-0.4143	

crease with increasing wave height, which implies less predictability as the sea state increases. This is intuitively satisfying.

- (iii) Most importantly, in every case, the Lyapunov spectrum exhibits the same structure: two positive exponents, third exponent close to zero, followed by two or three negative exponents (depending on the embedding dimension), with the sum of all Lyapunov exponents being negative.

Table X corresponds to the sea clutter data set from Argentia, Newfoundland. Because of the presence of noise in this case, the raw data show more than two positive Lyapunov exponents, whereas the filtered data behave in a similar way to that described above (with two positive exponents, the third one approximately zero, followed by negative exponents). For example, we observe that the Lyapunov spectrum of the raw data corresponding to range 5625 m has 10 exponents with 7 of them being positive and with a positive sum, and the BBA algorithm did not even compute the

Kaplan-Yorke dimension (see Table V). The reason for this non-chaotic behavior is the presence of a significant amount of noise in the sea clutter data.

Figure 15 illustrates the Lyapunov exponents for a typical FIR-filtered sea clutter data set collected during the experiment at Argentia, Newfoundland. The Lyapunov exponents are shown plotted as a function of L , denoting the number of times for which the linear terms in the Jacobian matrices used in the BBA algorithm are multiplied together. The exponents have been averaged over 3000 different starting locations on the attractor. We observe from this figure that there are two positive exponents and two negative exponents with the third exponent approaching zero. For this computation, we used an order of fit (for the polynomial) equal to 2.

The important conclusions drawn from the Lyapunov spectrum analysis of sea clutter (computed using the BBA algorithm) may be summarized as follows:

- (i) In all the cases (except for the noticeably noisy ones),

TABLE VIII. Lyapunov exponents of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set III, Dartmouth).

Range (m)	Lyapunov exponents (nats/sample)								
	Amplitude			In-phase component			Quadrature-phase component		
	Corrected	3 point	FIR	Corrected	3 point	FIR	Corrected	3 point	FIR
2739	+0.2370	+0.2616	+0.2965	+0.3943	+0.3162	+0.3895	+0.4064	+0.3196	+0.3994
	+0.1386	+0.1437	+0.1468	+0.2262	+0.1603	+0.2378	+0.2253	+0.1590	+0.2299
	+0.0229	-0.0218	-0.0012	+0.0574	+0.0088	+0.0392	+0.0450	-0.0004	+0.0437
	-0.1097	-0.2409	-0.2300	-0.2037	-0.1891	-0.2006	-0.1905	-0.2078	-0.2030
	-0.3125	-0.7274	-0.7851	-0.7684	-0.7264	-0.7896	-0.7847	-0.7227	-0.7778
	-0.8067								
2799	+0.3170	+0.3004	+0.3159	+0.4667	+0.3272	+0.4614	+0.4715	+0.3601	+0.2630
	+0.1658	+0.1495	+0.1743	+0.2811	+0.2008	+0.2743	+0.2787	+0.2149	+0.1184
	-0.0052	-0.0105	-0.0015	+0.0818	+0.0204	+0.0767	+0.1002	+0.0233	-0.0116
	-0.2799	-0.2591	-0.2594	-0.2287	-0.2058	-0.2276	-0.1974	-0.2057	-0.1876
	-0.8664	-0.8109	-0.8024	-0.8431	-0.7753	-0.8465	-0.8527	-0.8117	-0.7120
2859	+0.3421	+0.2841	+0.3240	+0.4256	+0.4902	+0.4102	+0.4288	+0.5096	+0.4186
	+0.1848	+0.1629	+0.1855	+0.2465	+0.2939	+0.2347	+0.2396	+0.3106	+0.2456
	+0.0009	-0.0113	-0.0085	+0.0566	+0.0920	+0.0629	+0.0563	+0.0858	+0.0538
	-0.2770	-0.2412	-0.2556	-0.1800	-0.1849	-0.1729	-0.1891	-0.1927	-0.1885
	-0.8557	-0.8206	-0.8399	-0.7609	-0.8063	-0.7353	-0.7509	-0.8028	-0.7279
2919	+0.3401	+0.3059	+0.3496	+0.4362	+0.3331	+0.4296	+0.5070	+0.3429	+0.4381
	+0.1922	+0.1541	+0.1857	+0.2787	+0.1494	+0.2796	+0.3009	+0.1509	+0.2716
	+0.0037	+0.0041	+0.0077	+0.1342	+0.0193	+0.1408	+0.0889	+0.0154	+0.1377
	-0.2401	-0.2413	-0.2589	-0.0235	-0.1522	-0.0235	-0.1637	-0.1739	-0.0218
	-0.8310	-0.8070	-0.8624	-0.2690	-0.6731	-0.2648	-0.7509	-0.6880	-0.2589
				-0.8157		-0.8026			-0.8144
2979	+0.3277	+0.3125	+0.3263	+0.4672	+0.3216	+0.4627	+0.4603	+0.3140	+0.4742
	+0.1750	+0.1782	+0.1966	+0.3171	+0.1441	+0.2672	+0.2711	+0.1343	+0.2643
	-0.0019	-0.0058	+0.0070	+0.1492	+0.0095	+0.0823	+0.0803	+0.0085	+0.0844
	-0.2313	-0.2195	-0.2334	+0.0136	-0.1553	-0.1421	-0.1592	-0.1740	-0.1617
	-0.7808	-0.7203	-0.8320	-0.2182	-0.6162	-0.7008	-0.6808	-0.6641	-0.6847
				-0.7701					

the algorithm invariably produced two positive exponents, which is further evidence of chaotic dynamics of sea clutter.

- (ii) The third exponent goes to zero consistently, indicating that sea clutter can be modelled as a set of coupled nonlinear differential equations.
- (iii) The sum of all exponents is negative, indicating a dissipative dynamical and hence a physically realizable system.

D. Kaplan-Yorke dimension

The Lyapunov dimension or the Kaplan-Yorke dimension for sea clutter was estimated from the Lyapunov exponents, using equation (19). This is conjectured to be the same as the fractal dimension estimator called the information dimension. The results presented in Tables I–V show that it did indeed agree very well with the estimated D_2 . This is yet another remarkable piece of evidence for the chaotic dynamics of sea clutter. However, due to the unavoidable presence of noise (even after filtering), the equality of D_2 and D_{KY} cannot always be guaranteed. Furthermore, the different algorithms used for the estimation of D_2 and the Lyapunov spectrum (from which D_{KY} is estimated) may not behave in identical ways with respect to the presence of noise in the

preprocessed clutter data. It is therefore not surprising to see that D_2 is sometimes slightly greater than D_{KY} , and some other times the other way round. (According to results on the Hénon map, Kaplan and Yorke map, and Zaslavskii map reported in Russell *et al.*,²⁹ a similar behavior concerning the estimated values of D_2 and D_{KY} could be observed.) However, the important point to observe here is that D_2 and D_{KY} are consistently very close to each other.

X. DISCUSSION

A. Summary of experimental results and conclusions

The experimental study reported in this paper has clearly demonstrated the chaotic dynamics of sea clutter in the most complete manner reported in the literature to date. This conclusion is based on the results of different radar experiments, data from which were used to compute the correlation dimension, Lyapunov exponents, and Kaplan-Yorke dimension. The results are summarized here:

- (i) The correlation dimension D_2 of sea clutter lies in the range 4.1 to 4.5, independently of time of the day, month of the year, radar range and range resolution, type of like-polarization, sea state, radar location, and radar type. It is also essentially the same regardless of

TABLE IX. Lyapunov exponents of in-phase, quadrature-phase and amplitude of clutter data for different ranges (data set IV, Cape Bonavista).

Range (m)	Lyapunov exponents (nats/sample)								
	Amplitude			In-phase component			Quadrature-phase component		
	Corrected	3 point	FIR	Corrected	3 point	FIR	Corrected	3 point	FIR
5730	+0.5052	+0.2504	+0.2389	+0.3736	+0.1955	+0.1943	+0.2827	+0.1903	+0.1941
	+0.3376	+0.1184	+0.1185	+0.2209	+0.0941	+0.0829	+0.1662	+0.0816	+0.0795
	+0.1157	-0.0269	-0.0191	+0.0516	-0.0285	-0.0272	+0.0457	-0.0309	-0.0102
	-0.1997	-0.2599	-0.2214	-0.2492	-0.1730	-0.1376	-0.1264	-0.1852	-0.1441
	-0.8868	-0.7601	-0.6817	-0.8664	-0.5607	-0.4976	-0.3714	-0.5809	-0.5299
5850							-0.9841		
	+0.5686	+0.3420	+0.2902	+0.4722	+0.2186	+0.1922	+0.3354	+0.2108	+0.1927
	+0.4080	+0.1826	+0.1501	+0.3155	+0.1011	+0.0808	+0.2078	+0.1111	+0.0843
	+0.2294	-0.0111	-0.0181	+0.1176	-0.0307	-0.0189	+0.0672	-0.0304	-0.0165
	-0.0128	-0.2874	-0.2488	-0.2071	-0.2403	-0.1593	-0.1230	-0.2049	-0.1518
6000	-0.3428	-0.8999	-0.8286	-0.8619	-0.6926	-0.5912	-0.4193	-0.6722	-0.5990
	-1.0730						-1.0303		
	+0.5032	+0.2591	+0.2278	+0.4204	+0.2713	+0.2519	+0.4221	+0.2625	+0.3060
	+0.3209	+0.1215	+0.1157	+0.2439	+0.1305	+0.1298	+0.2516	+0.1368	+0.1682
	+0.1166	-0.0239	-0.0179	+0.0712	-0.0257	-0.0111	+0.0579	-0.0056	-0.0032
6150	-0.2283	-0.2645	-0.2200	-0.2562	-0.2371	-0.1932	-0.2215	-0.2115	-0.2073
	-0.8903	-0.7352	-0.6528	-0.8767	-0.7275	-0.5884	-0.8562	-0.6925	-0.6825
	+0.2856	+0.2593	+0.2603	+0.3696	+0.2712	+0.2384	+0.3812	+0.3786	+0.3282
	+0.1700	+0.1244	+0.1351	+0.2039	+0.1493	+0.1164	+0.2501	+0.2149	+0.1608
	+0.0445	-0.0219	-0.0162	+0.0346	-0.0065	+0.0021	+0.1023	+0.0429	+0.0227
6300	-0.1343	-0.2275	-0.2395	-0.2295	-0.1924	-0.1596	-0.0749	-0.2310	-0.1821
	-0.3930	-0.7168	-0.6800	-0.8486	-0.6921	-0.5758	-0.3669	-0.7797	-0.6472
	-0.9669						-0.9662		
	+0.5427	+0.2676	+0.2369	+0.4657	+0.2532	+0.1956	+0.4263	+0.2658	+0.2594
	+0.3659	+0.1382	+0.1151	+0.2783	+0.1441	+0.0765	+0.2543	+0.1251	+0.1391
6450	+0.1458	-0.0143	-0.0285	+0.0789	-0.0056	-0.0175	+0.0485	-0.0134	-0.0058
	-0.1774	-0.2573	-0.2250	-0.2195	-0.1914	-0.1362	-0.2649	-0.2308	-0.1829
	-0.8889	-0.8092	-0.7069	-0.8607	-0.6604	-0.5476	-0.9448	-0.7071	-0.6651
	+0.4192	+0.2550	+0.2644	+0.3566	+0.1670	+0.1377	+0.3222	+0.1610	+0.1401
	+0.2673	+0.1347	+0.1376	+0.2006	+0.0704	+0.0386	+0.1784	+0.0660	+0.0566
6600	+0.0507	-0.0209	-0.0152	+0.0035	-0.0296	-0.0099	+0.0118	-0.0324	-0.0186
	-0.2269	-0.2588	-0.2216	-0.2336	-0.1702	-0.1126	-0.2337	-0.1505	-0.1185
	-0.8992	-0.7493	-0.6773	-0.8522	-0.5251	-0.4304	-0.7957	-0.5450	-0.4426
	+0.3368	+0.2241	+0.2323	+0.2718	+0.1660	+0.1143	+0.2631	+0.1791	+0.1847
	+0.1996	+0.1058	+0.1082	+0.1369	+0.0695	+0.0406	+0.1326	+0.0696	+0.0650
6600	+0.0240	-0.0165	-0.0182	-0.0331	-0.0230	-0.0206	-0.0194	-0.0301	-0.0184
	-0.2856	-0.2305	-0.2265	-0.2281	-0.1565	-0.1127	-0.2396	-0.1680	-0.1466
	-0.8960	-0.6690	-0.6473	-0.7430	-0.4979	-0.3995	-0.7670	-0.5300	-0.5456

whether the radar component used in the computation is the in-phase, quadrature-phase, or amplitude component.

- (ii) The global embedding dimension d_E is either 5 or 6.
- (iii) The local embedding dimension (dynamical embedding dimension) d_L is either 5 or 6. This shows that the number of true Lyapunov exponents is either 5 or 6, with $d_L \leq d_E$.
- (iv) The Lyapunov spectrum consists of two positive exponents, followed by one exponent equal to zero (within experimental errors), and two or three negative exponents. This composition of the Lyapunov spectrum is independent of both radar range and the radar component being considered. Moreover, the dis-

tribution of the Lyapunov exponents is independent of sea state. However, their absolute values do depend on sea state and radar type.

- (v) The Kaplan-Yorke dimension, a byproduct of the Lyapunov spectrum, is consistently very close to D_2 .

These results were obtained using carefully ground-truthed radar data collected with an instrument-quality radar at two different locations, and a commercial marine radar at another location. The data were calibrated as appropriate, and pre-processed to minimize the effects of noise and roundoff errors, always taking extra care to protect the integrity of the actual clutter process.

In an article¹⁴ published in 1990, preliminary results ob-

TABLE X. Lyapunov exponents amplitude of clutter data for different ranges (data set V, Argentina).

Range (m)	Lyapunov exponents (nats/sample)		
	Raw	3 point	FIR
4500	+0.5123	+0.3175	+0.2435
	+0.3595	+0.1695	+0.1128
	+0.1759	-0.0060	-0.0295
	-0.0305	-0.2580	-0.2305
	-0.3464	-0.8652	-0.7008
	-1.0701		
5625	+0.7668	+0.5332	+0.3543
	+0.6627	+0.3622	+0.1452
	+0.5619	+0.1402	-0.0052
	+0.4483	-0.1780	-0.1921
	+0.3455	-0.8994	-0.7365
	+0.2149		
6900	+0.0541		
	-0.1719		
	-0.4825		
	-1.2132		
	+0.5267	+0.3349	+0.2848
	+0.3982	+0.1980	+0.1485
	+0.2608	+0.0295	+0.0032
	+0.1118	-0.2332	-0.2229
	-0.1095	-0.8751	-0.7926
	-0.4201		
	-1.0938		

tained using the Grassberger-Procaccia algorithm were presented, in light of which we posed the question:

Is there a radar clutter attractor?

On the basis of the many positive results presented in this paper, we are now emboldened to answer this question with an emphatic “yes.”

Furthermore, the various parameter estimation algorithms used in this paper have revealed highly significant information about the chaotic dynamics of sea clutter, as concluded here:

- (1) Sea clutter, as observed by a microwave radar (regardless of whether it is coherent or noncoherent) at low grazing angles, is generated by a low degree-of-freedom strange attractor. The small value of correlation dimension D_2 , lying between 4.1 and 4.5, for varying environmental conditions is especially interesting since the intrinsic dissipation is relatively weak.
- (2) The results of Lyapunov spectrum analysis reveal that the generation of sea clutter is governed by a coupled system of five or six nonlinear differential equations: this is a remarkable result deduced from the fact that the third Lyapunov exponent is consistently zero (within numerical errors) for an embedding dimension of 5 or 6. However, the derivation of these equations requires that we go back to the basic physical principles that govern the generation of sea clutter. This is indeed a difficult research topic, which is most deserving of detailed attention.

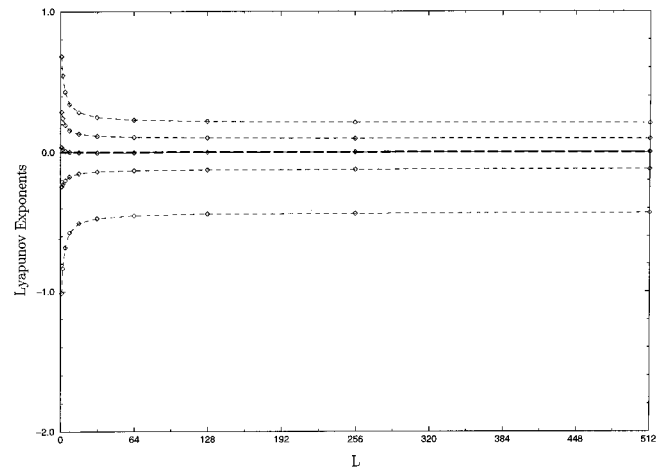


FIG. 15. Lyapunov exponents for a typical FIR-filtered sea clutter data set.

- (3) The correlation dimension and Kaplan-Yorke dimension are relatively insensitive to variations in sea state. In contrast, the absolute values of Lyapunov exponents, for a given range and radar system, do exhibit significant variations with respect to sea state. This latter result demonstrates how the behavior of a chaotic attractor can vary within a fixed dimensionality. The result is also reassuring as it emphasizes the non-stationary nature of sea clutter observed over a long period of time. It should, however, be noted that for each sea state, the Lyapunov spectrum analysis was performed using a set of sea clutter data short enough to satisfy stationarity of the data as required by the estimation algorithms and yet long enough to produce reliable results.

B. Supporting physical evidence

There is also strong supporting physical evidence for our research findings, as indicated in what follows.

Sea clutter is naturally related to fluid turbulence in an ocean environment. In 1971, Ruelle and Takens¹⁰⁸ proposed a chaotic mechanism for the generation of turbulence in a fluid and related phenomena in a dissipative system. The existence of chaos in fluid turbulence was subsequently confirmed by Abarbanel *et al.*¹⁰⁹ Indeed, the low value of D_2 or D_{KY} in the range of 4.1 to 4.5 reported in this paper for sea clutter is typical for fluid-related phenomena.

In Section II, we discussed the role of gravity waves in the generation of sea clutter. In this context, the results of chaotic analysis performed by Frison and Abarbanel¹¹⁰ on ocean gravity waves measured at the Harvest platform off the coast of California are highly relevant. They reported a $D_{KY} \approx 4.5$, which is essentially the same as that for sea clutter reported herein. The value of D_{KY} reported by Frison and Abarbanel is computed from a Lyapunov spectrum for ocean gravity waves that is identical in structure to that of sea clutter.

C. Dynamic reconstruction

The following quote, taken from the preface to a book by Gray and Davisson¹¹¹ on random processes, is ever so true in the case of sea clutter:

“One can argue that given complete knowledge of the physics of an experiment, the outcome must always be predictable, at least with the aid of a sufficiently powerful computer.”

Although we do not have knowledge of the system of nonlinear differential equations that describes the physical generation of sea clutter, the research findings reported in this paper point to the existence of such a system consisting of 5 or 6 equations. This observation, in turn, encourages us to invoke two important theorems:

- (1) **Delay Embedding Theorem**,⁷³ which states that the underlying nonlinear dynamics of a physical (chaotic) phenomenon can be reconstructed in the form of a predictive model by using a time series based on one variable of the phenomenon; and
- (2) **Universal Approximation Theorem**,¹¹² which teaches us that given data representative of a physical (chaotic) phenomenon, it is possible to construct a neural network (in the form of a multilayer perceptron or radial-basis function network) that approximates the predictive model.

Thus, despite the lack of knowledge of the exact physical laws describing the generation of sea clutter, we are assured by these two theorems that we can indeed build a predictive model capable of capturing the underlying nonlinear dynamics of sea clutter. The construction of this predictive model in the form of a regularized radial-basis function network for sea clutter under varying conditions is described in a forthcoming paper.¹¹³ The results presented therein provide further proof for the chaotic dynamics of sea clutter.

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APPENDIX: LIST OF ABBREVIATIONS

BBA	:	Brown, Bryant and Abarbanel
D_2	:	Correlation dimension
D_{KY}	:	Kaplan-Yorke dimension
D_{ML}	:	Maximum likelihood estimate of correlation dimension
FFT	:	Fast Fourier transform
FIR	:	Finite impulse response
GPA	:	Grassberger-Procaccia algorithm
GFNN	:	Global false nearest neighbor
HH	:	Horizontal on both transmit and receive (polarization)
LFNN	:	Local false nearest neighbor
MI	:	Mutual information
PRF	:	Pulse repetition frequency
SCD	:	Surrogate data analysis using correlation dimension as the discriminating statistic
SIPD	:	Surrogate data analysis using the Inter-point distances as the discriminating statistic
SNR	:	Signal-to-noise ratio
STB	:	Schouten, Takens and Bleek
VTs	:	Vessel traffic system
VV	:	Vertical on both transmit and receive (polarization)
WSF	:	Wave-shaping filter
d_E	:	Embedding dimension
d_L	:	Dynamical dimension

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